DS_HW2_402101509

May 23, 2026

1 Regression Methods

1.0.1 Loading required materials

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import MinMaxScaler, StandardScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
from sklearn.linear_model import LinearRegression, LogisticRegression, Ridge,␣
 ↪Lasso
from sklearn.kernel_ridge import KernelRidge
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn import tree
from sklearn.metrics import RocCurveDisplay, auc, roc_curve
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier
from xgboost import XGBClassifier
from imblearn.over_sampling import SMOTE
import lightgbm as lgbm
from catboost import CatBoostClassifier
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    roc_curve,
    roc_auc_score,
    cohen_kappa_score,
    log_loss
)

# Disable warnings
import warnings
```

```
def warn(*args, **kwargs):
    pass
warnings.warn = warn

# Load data
odf = pd.read_csv("Hotel_Bookings_Clean.csv")    # We make an original version␣
 ↪of the dataset for later usage
odf.pop("Unnamed: 0")
df = odf.copy(True)
```

1.0.2 Preprocessing

[2]:
```
# ADR is undoubtedly a non negative value
df = df[df["adr"] >= 0]
# Next remove the  large outliers, but in a way that perserves 99.9% of the␣
 ↪data. This is to make sure deleted vaues are really outliers
df = df[df["adr"] <= df["adr"].quantile(.999)]
# Also exclude canceled and no-show reservations
df = df[(df["Canceled"] != 1) & (df["No-Show"] != 1)]

df["adr"].describe()
```

[2]:
```
count    74661.000000
mean        99.929436
std         48.411436
min          0.000000
25%         68.000000
50%         92.900000
75%        125.000000
max        326.330000
Name: adr, dtype: float64
```

1.0.3 Normalization

1.0.4 We Normalize the adr column and after finishing our analysis we will scale the values back up.

[3]:
```
scaler = MinMaxScaler()

# We keep an unnormalized version for later use
ndf = df.copy(deep=True)
ndf[ndf.columns] = scaler.fit_transform(ndf[ndf.columns])
ndf.describe()
```

[3]:
```
          Unnamed: 0     lead_time  arrival_date_year  arrival_date_month  \
count  74661.000000  74661.000000       74661.000000        74661.000000
mean       0.556235      0.109018           0.574095            0.499890
```

```
std        0.313692      0.123770           0.351637            0.318758
min        0.000000      0.000000           0.000000            0.000000
25%        0.253097      0.012212           0.500000            0.181818
50%        0.677558      0.062415           0.500000            0.545455
75%        0.843411      0.169607           1.000000            0.727273
max        1.000000      1.000000           1.000000            1.000000

       arrival_date_week_number  arrival_date_day_of_month  \
count              74661.000000               74661.000000
mean                   0.501562                   0.494710
std                    0.267003                   0.292495
min                    0.000000                   0.000000
25%                    0.288462                   0.233333
50%                    0.519231                   0.500000
75%                    0.711538                   0.733333
max                    1.000000                   1.000000

       stays_in_weekend_nights  stays_in_week_nights        adults  \
count             74661.000000          74661.000000  74661.000000
mean                  0.058167              0.060151      0.457943
std                   0.061843              0.046647      0.127204
min                   0.000000              0.000000      0.000000
25%                   0.000000              0.024390      0.500000
50%                   0.062500              0.048780      0.500000
75%                   0.125000              0.073171      0.500000
max                   1.000000              1.000000      1.000000

           children  …    No Deposit    Non Refund    Refundable  Canceled  \
count  74661.000000  …  74661.000000  74661.000000  74661.000000   74661.0
mean       0.033904  …      0.997067      0.001246      0.001688       0.0
std        0.129793  …      0.054080      0.035272      0.041046       0.0
min        0.000000  …      0.000000      0.000000      0.000000       0.0
25%        0.000000  …      1.000000      0.000000      0.000000       0.0
50%        0.000000  …      1.000000      0.000000      0.000000       0.0
75%        0.000000  …      1.000000      0.000000      0.000000       0.0
max        1.000000  …      1.000000      1.000000      1.000000       0.0

       Check-Out  No-Show   stay_nights        guests  avg_nationality_adr  \
count    74661.0  74661.0  74661.000000  74661.000000         74661.000000
mean         0.0      0.0      0.059594      0.161988             0.272397
std          0.0      0.0      0.044888      0.055569             0.044684
min          0.0      0.0      0.000000      0.000000             0.000000
25%          0.0      0.0      0.035088      0.166667             0.221706
50%          0.0      0.0      0.052632      0.166667             0.274825
75%          0.0      0.0      0.070175      0.166667             0.303330
max          0.0      0.0      1.000000      1.000000             1.000000
```

```
       avg_nationality_adr_enc
count             74661.000000
mean                  0.345956
std                   0.183252
min                   0.000000
25%                   0.154286
50%                   0.337143
75%                   0.462857
max                   1.000000

[8 rows x 41 columns]
```

1.0.5 Prepering train and test datasets

[4]:
```
ndfc = ndf.copy(True)
adr = ndfc.pop("adr")
X_train, X_test, y_train, y_test = train_test_split(ndfc,adr,test_size=0.7)
print(X_train.shape)
print(X_test.shape)
```

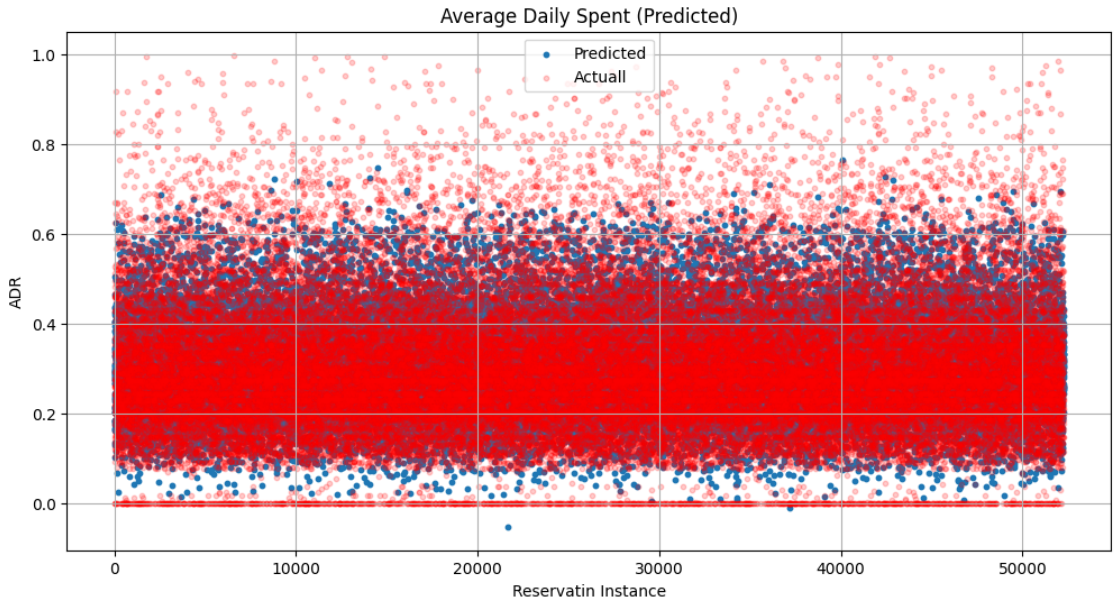
```
(22398, 40)
(52263, 40)
```

1.0.6 Applying a linear model on the unnormalized dataset

[5]:
```
model = LinearRegression()
model.fit(X_train, y_train)
```

[5]:
```
LinearRegression()
```

[6]:
```
predicted_adr_train = model.predict(X_train)
predicted_adr_test = model.predict(X_test)
x = np.arange(0,len(predicted_adr_test),1)
plt.figure(figsize=(12,6))
plt.scatter(x,predicted_adr_test,s=10)
plt.scatter(x,y_test,alpha=.2,s=10,c="r")
plt.legend(["Predicted","Actuall"])
plt.grid()
plt.title("Average Daily Spent (Predicted)")
plt.ylabel("ADR")
plt.xlabel("Reservatin Instance")
plt.show()
```



1.0.7 Measuring different error kinds

1.0.8 Note that since there are zeros among our actuall data, we can not use MAPE as it will result in very high and misleading numbers.

[7]:
```
print('RMSE on the train set is: %.2f'␣
 ↪%pow(mean_squared_error(y_train,predicted_adr_train),.5))
print('RMSE on the test set is: %.2f'␣
 ↪%pow(mean_squared_error(y_test,predicted_adr_test),0.5))
print('MAE on the train set is %.2f' %mean_absolute_error(y_train,␣
 ↪predicted_adr_train))
print('MAE on the test set is %.2f' %mean_absolute_error(y_test,␣
 ↪predicted_adr_test))
print('R2 Score is %.3f' %r2_score(y_test, predicted_adr_test))
```

```
RMSE on the train set is: 0.11
RMSE on the test set is: 0.12
MAE on the train set is 0.09
MAE on the test set is 0.09
R2 Score is 0.393
```

We see that all errors are relatively high even for train dataset. The reason is that the linear model can not predict zero inflated (many zeros among many positive values) data as it is shown in the plot. Not a single zero value has been predicted.

There are workarounds to this problem that still uses linear regression, but for the sake of practice we are going to use another model.

1.0.9 Kernel Regression

Since the whole dataset is to large to apply kernel on, we use a subset of the data for our analysis

[8]:
```
kdf = ndf[:10000]
kadr = kdf.pop("adr")
X_train_kernel, X_test_kernel, y_train_kernel, y_test_kernel =␣
 ↪train_test_split(kdf,kadr,test_size=0.7)

model = KernelRidge(kernel="sigmoid")
model.fit(X_train_kernel, y_train_kernel)

predicted_adr_train = model.predict(X_train_kernel)
predicted_adr_test = model.predict(X_test_kernel)

print('RMSE on the train set is: %.2f'␣
 ↪%pow(mean_squared_error(y_train_kernel,predicted_adr_train),.5))
print('RMSE on the test set is: %.2f'␣
 ↪%pow(mean_squared_error(y_test_kernel,predicted_adr_test),0.5))
print('MAE on the train set is %.2f' %mean_absolute_error(y_train_kernel,␣
 ↪predicted_adr_train))
print('MAE on the test set is %.2f' %mean_absolute_error(y_test_kernel,␣
 ↪predicted_adr_test))
print('R2 Score is %.3f' %r2_score(y_test_kernel, predicted_adr_test))
```

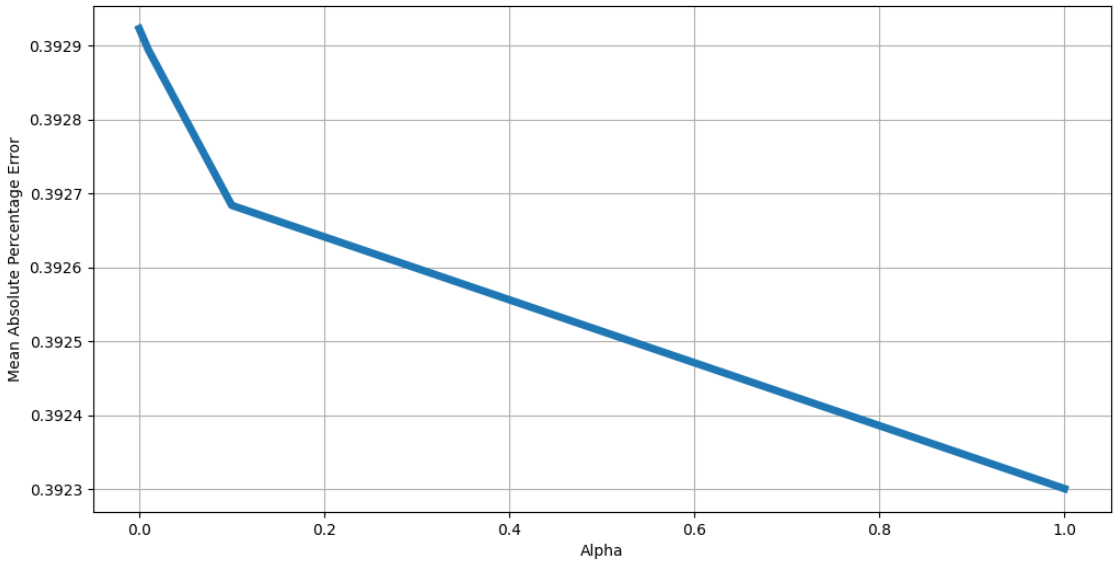
```
RMSE on the train set is: 0.12
RMSE on the test set is: 0.12
MAE on the train set is 0.09
MAE on the test set is 0.09
R2 Score is 0.340
```

Through trial and error it is visible that sigmoid kernel results in the best scores, but still it is not very much better than linear model. Even increasing the size of the portion that the analysis is done on, results in convergance of all metrics.

1.0.10 Ridge

[9]:
```
alpha_list = [0.000001,.0001,.0005,.001,.005,0.01,.1,1]
scores = []
for alpha in alpha_list:
  model = Ridge(alpha=alpha)
  model.fit(X_train, y_train)
  predicted_prices_test = model.predict(X_test)
  MAPE_test = r2_score(y_test, predicted_prices_test)
  scores.append(MAPE_test)
plt.figure(figsize=(12,6))
plt.plot(alpha_list,scores,lw=5)
plt.xlabel('Alpha')
```

```
plt.ylabel('Mean Absolute Percentage Error')
plt.grid()
plt.show()
```



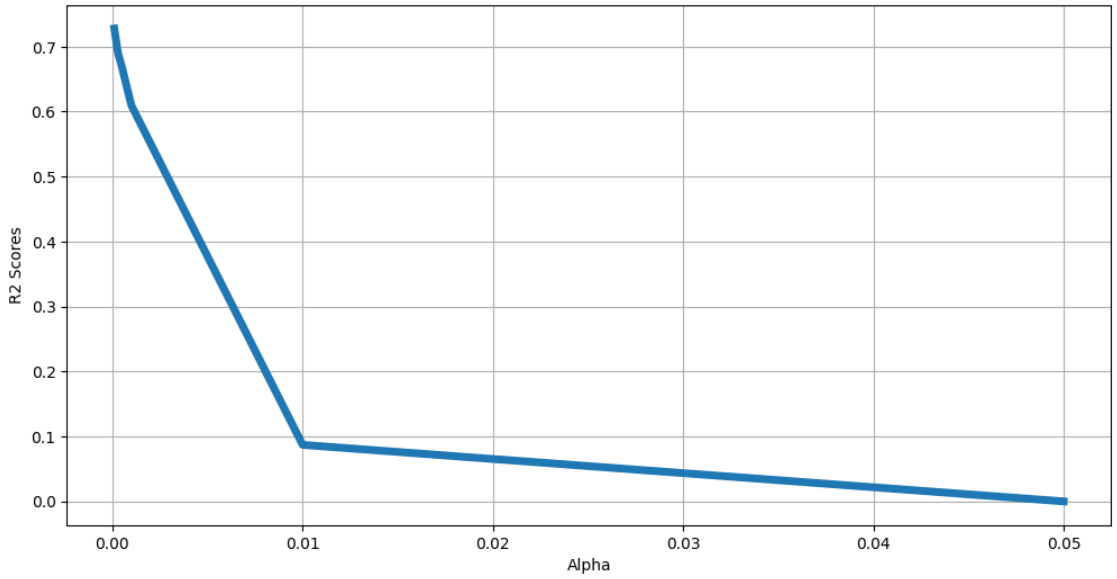
Ridge regression is almost the same as linear in r2 metric.

1.0.11 Lasso Regression

[10]:
```
from sklearn.preprocessing import PolynomialFeatures
from sklearn.feature_selection import SelectKBest, mutual_info_regression

pdf = ndf[:10000].copy(True)
padr = pdf.pop('adr')
X_train_poly, X_test_poly, y_train_poly, y_test_poly =␣
 ↪train_test_split(pdf,padr,test_size=0.7)

poly = PolynomialFeatures(degree=2)
X_train_transformed_poly = poly.fit_transform(X_train_poly)
X_test_transformed_poly = poly.transform(X_test_poly)

feature_selector = SelectKBest(score_func=mutual_info_regression, k = 'all').
 ↪fit(X_train_transformed_poly,y_train_poly)

scores = []
selector = SelectKBest(mutual_info_regression, k = 400)
selector.fit(X_train_transformed_poly, y_train_poly)
X_train_transformed = selector.transform(X_train_transformed_poly)
```

```
X_test_transform = selector.transform(X_test_transformed_poly)

alpha_list = [.0001,0.0003,0.0005,.001,0.01,0.05]
scores = []
for alpha in alpha_list:
    model = Lasso(alpha=alpha)
    model.fit(X_train_transformed, y_train_poly)
    predicted_prices_test = model.predict(X_test_transform)
    MAPE_test = r2_score(y_test_poly, predicted_prices_test)
    scores.append(MAPE_test)

plt.figure(figsize=(12,6))
plt.plot(alpha_list,scores,lw=5)
plt.xlabel('Alpha')
plt.ylabel('R2 Scores')
plt.grid()
plt.show()
```

```
/opt/anaconda3/lib/python3.12/site-
packages/sklearn/linear_model/_coordinate_descent.py:678: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.764e-01, tolerance: 6.810e-03
  model = cd_fast.enet_coordinate_descent(
```



Lasso regression results in a better R2 score than the other methods.

1.0.12 Elastic Net

[11]:
```
from sklearn.linear_model import ElasticNet

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

elastic_net = ElasticNet(alpha=.001, l1_ratio=.7, random_state=42)
elastic_net.fit(X_train_scaled, y_train)

y_pred_train = elastic_net.predict(X_train_scaled)
y_pred_test = elastic_net.predict(X_test_scaled)

print(f"R2: {r2_score(y_train, y_pred_train):.4f}")
print(f"R2: {r2_score(y_test, y_pred_test):.4f}")
print(f"MAE:{mean_absolute_error(y_test, y_pred_test):.2f}")
```

```
R2: 0.4005
R2: 0.3914
MAE:0.09
```

1.0.13 Kernel Trick:

1.0.14 What is the best model: According R2 metric, the best model (with some tuning) is the Lasso regressor.

1.0.15 When to use each model:

1.1 Binary Classification Methods

We try to use binary classification to determine the status of the reservation as a binary classifier (i.e. canceled or not). Later we use multi class classification (i.e. canceled, checked out or no show) and compare the two approach.

1.1.1 Logistic Regression

[ ]:
```
lgr_df = odf.copy(True)
canceles = lgr_df.pop("Canceled")
no_show = lgr_df.pop("No-Show")
check_out = lgr_df.pop("Check-Out")
X_train, X_test, y_train, y_test = train_test_split(lgr_df,canceles,test_size=0.
↪7)

model = LogisticRegression(penalty='l1',solver="liblinear",max_iter=1000)
model.fit(X_train, y_train)
predictions_train = model.predict(X_train)
predictions_test = model.predict(X_test)

errors_train = (predictions_train != y_train).sum()
```

```python
errors_test = (predictions_test != y_test).sum()

print("="*50)
print("TRAINING SET METRICS")
print("="*50)
print(f"Accuracy: {accuracy_score(y_train, predictions_train):.4f}")
print(f"Precision: {precision_score(y_train, predictions_train):.4f}")
print(f"Recall: {recall_score(y_train, predictions_train):.4f}")
print(f"F1-Score: {f1_score(y_train, predictions_train):.4f}")
print(f"AUC: {roc_auc_score(y_train, model.predict_proba(X_train)[:, 1]):.4f}")
print("\nConfusion Matrix:")
print(confusion_matrix(y_train, predictions_train))

print("\n" + "="*50)
print("TEST SET METRICS")
print("="*50)
print(f"Accuracy: {accuracy_score(y_test, predictions_test):.4f}")
print(f"Precision: {precision_score(y_test, predictions_test):.4f}")
print(f"Recall: {recall_score(y_test, predictions_test):.4f}")
print(f"F1-Score: {f1_score(y_test, predictions_test):.4f}")
print(f"AUC: {roc_auc_score(y_test, model.predict_proba(X_test)[:, 1]):.4f}")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, predictions_test))
```

```
==================================================
TRAINING SET METRICS
==================================================
Accuracy: 0.9338
Precision: 0.8876
Recall: 0.9357
F1-Score: 0.9110
AUC: 0.9774

Confusion Matrix:
[[21224  1530]
 [  830 12085]]

==================================================
TEST SET METRICS
==================================================
Accuracy: 0.9343
Precision: 0.8873
Recall: 0.9369
F1-Score: 0.9114
AUC: 0.9767

Confusion Matrix:
[[49619  3575]
```

```
 [ 1895 28140]]
```

1.1.2 Support Vector Machine

[ ]:
```
svmdf = odf.copy(True)

canceles = svmdf.pop("Canceled")
no_show = svmdf.pop("No-Show")
check_out = svmdf.pop("Check-Out")

X_train, X_test, y_train, y_test = train_test_split(svmdf, canceles,␣
 ↪test_size=0.3)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = SVC(
    C=3,
    kernel='linear',
    gamma='scale',
    class_weight='balanced',
    probability=True,
    max_iter=5000,
)

model.fit(X_train, y_train)

t = 0.45
probs_train = model.predict_proba(X_train)[:, 1]
predictions_train = (probs_train > t).astype(int)
probs_test = model.predict_proba(X_test)[:, 1]
predictions_test = (probs_test > t).astype(int)

print("="*50)
print("TRAINING SET METRICS")
print("="*50)
print(f"Accuracy: {accuracy_score(y_train, predictions_train):.4f}")
print(f"Precision: {precision_score(y_train, predictions_train):.4f}")
print(f"Recall: {recall_score(y_train, predictions_train):.4f}")
print(f"F1-Score: {f1_score(y_train, predictions_train):.4f}")
print("\nConfusion Matrix:")
print(confusion_matrix(y_train, predictions_train))

print("\n" + "="*50)
print("TEST SET METRICS")
print("="*50)
print(f"Accuracy: {accuracy_score(y_test, predictions_test):.4f}")
```

```
print(f"Precision: {precision_score(y_test, predictions_test):.4f}")
print(f"Recall: {recall_score(y_test, predictions_test):.4f}")
print(f"F1-Score: {f1_score(y_test, predictions_test):.4f}")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, predictions_test))

weights_linear = model.coef_[0]
fig, ax = plt.subplots(figsize=(20,8))
y_pos = np.arange(0,len(weights_linear),1)
labels = list(svmdf.columns)
hbars = ax.barh(y_pos, weights_linear, align='center')
ax.set_yticks(y_pos)
ax.set_yticklabels(labels)
ax.invert_yaxis()  # labels read top-to-bottom
ax.set_xlabel('Weights')
ax.set_title("Importance of each column from the algorithm's point of view")
plt.show()
```

```
/opt/anaconda3/lib/python3.12/site-packages/sklearn/svm/_base.py:297:
ConvergenceWarning: Solver terminated early (max_iter=5000).  Consider pre-
processing your data with StandardScaler or MinMaxScaler.
  warnings.warn(

==================================================
TRAINING SET METRICS
==================================================
Accuracy: 0.8972
Precision: 0.8697
Recall: 0.8420
F1-Score: 0.8557

Confusion Matrix:
[[49313  3798]
 [ 4758 25359]]

==================================================
TEST SET METRICS
==================================================
Accuracy: 0.8958
Precision: 0.8633
Recall: 0.8440
F1-Score: 0.8535

Confusion Matrix:
[[21122  1715]
 [ 2002 10831]]
```

1.1.3 Kernel SVM

[ ]:
```
lgrdf = odf.copy(True)

canceles = lgrdf.pop("Canceled")
no_show = lgrdf.pop("No-Show")
check_out = lgrdf.pop("Check-Out")

X_train, X_test, y_train, y_test = train_test_split(lgrdf, canceles,␣
 ↪test_size=0.3)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = SVC(
    C=3,
    kernel='rbf',
    gamma='scale',
    class_weight='balanced',
    probability=True,
    max_iter=5000,
)

model.fit(X_train, y_train)

for t in [0.3, 0.4, 0.5, 0.6, 0.7]:
    probs_train = model.predict_proba(X_train)[:, 1]
    predictions_train = (probs_train > t).astype(int)

    probs_test = model.predict_proba(X_test)[:, 1]
    predictions_test = (probs_test > t).astype(int)

    tn, fp, fn, tp = confusion_matrix(y_train, predictions_train).ravel()
    print(f"Train Threshold={t} | FP={fp}, FN={fn}")

    tn, fp, fn, tp = confusion_matrix(y_test, predictions_test).ravel()
    print(f"Test Threshold={t} | FP={fp}, FN={fn}")

print("="*50)
print("TRAINING SET METRICS")
print("="*50)
print(f"Accuracy: {accuracy_score(y_train, predictions_train):.4f}")
print(f"Precision: {precision_score(y_train, predictions_train):.4f}")
print(f"Recall: {recall_score(y_train, predictions_train):.4f}")
print(f"F1-Score: {f1_score(y_train, predictions_train):.4f}")
print("\nConfusion Matrix:")
print(confusion_matrix(y_train, predictions_train))
```

```python
print("\n" + "="*50)
print("TEST SET METRICS")
print("="*50)
print(f"Accuracy: {accuracy_score(y_test, predictions_test):.4f}")
print(f"Precision: {precision_score(y_test, predictions_test):.4f}")
print(f"Recall: {recall_score(y_test, predictions_test):.4f}")
print(f"F1-Score: {f1_score(y_test, predictions_test):.4f}")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, predictions_test))
```

```
/opt/anaconda3/lib/python3.12/site-packages/sklearn/svm/_base.py:297:
ConvergenceWarning: Solver terminated early (max_iter=5000).  Consider pre-
processing your data with StandardScaler or MinMaxScaler.
  warnings.warn(

Train Threshold=0.3 | FP=1568, FN=219
Test Threshold=0.3 | FP=748, FN=220
Train Threshold=0.4 | FP=1407, FN=272
Test Threshold=0.4 | FP=650, FN=266
Train Threshold=0.5 | FP=1232, FN=369
Test Threshold=0.5 | FP=564, FN=353
Train Threshold=0.6 | FP=1076, FN=529
Test Threshold=0.6 | FP=483, FN=447
Train Threshold=0.7 | FP=904, FN=904
Test Threshold=0.7 | FP=408, FN=612
==================================================
TRAINING SET METRICS
==================================================
Accuracy: 0.9783
Precision: 0.9699
Recall: 0.9699
F1-Score: 0.9699

Confusion Matrix:
[[52312   904]
 [  904 29108]]

==================================================
TEST SET METRICS
==================================================
Accuracy: 0.9714
Precision: 0.9680
Recall: 0.9527
F1-Score: 0.9603

Confusion Matrix:
[[22324   408]
```

```
[  612 12326]]
```

Kernel SVM resulted in pretty good metrics but it is orders of magnitude slower and computationally heavier than other methods so it is not really preferable.

1.1.4 KNN

[ ]:
```
knndf = odf.copy(True)

canceles = knndf.pop("Canceled")
no_show = knndf.pop("No-Show")
check_out = knndf.pop("Check-Out")

X_train, X_test, y_train, y_test = train_test_split(knndf, canceles,␣
 ↪test_size=0.3, random_state=42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Define model first
model = KNeighborsClassifier(n_neighbors=6, weights='uniform')
model.fit(X_train, y_train)

predictions_train = model.predict(X_train)
predictions_test = model.predict(X_test)

# print('Macro F1-score on Training set is %.2f' % f1_score(y_train,␣
 ↪predictions_train, average='macro'))
# print('Macro F1-score on Test set is %.2f' % f1_score(y_test,␣
 ↪predictions_test, average='macro'))

# print("="*50)
# print("TRAINING SET METRICS")
# print("="*50)
# print(f"Accuracy: {accuracy_score(y_train, predictions_train):.4f}")
# print(f"Precision: {precision_score(y_train, predictions_train):.4f}")
# print(f"Recall: {recall_score(y_train, predictions_train):.4f}")
# print(f"F1-Score: {f1_score(y_train, predictions_train):.4f}")
# print("\nConfusion Matrix:")
# print(confusion_matrix(y_train, predictions_train))

# print("\n" + "="*50)
# print("TEST SET METRICS")
# print("="*50)
# print(f"Accuracy: {accuracy_score(y_test, predictions_test):.4f}")
# print(f"Precision: {precision_score(y_test, predictions_test):.4f}")
# print(f"Recall: {recall_score(y_test, predictions_test):.4f}")
```

```
# print(f"F1-Score: {f1_score(y_test, predictions_test):.4f}")
# print("\nConfusion Matrix:")
# print(confusion_matrix(y_test, predictions_test))

# # Tuning neighbors
# neighbors_list = [3, 5, 7, 9, 15]
# scores = []
# scores_train = []

# for n in neighbors_list:
#     model = KNeighborsClassifier(n_neighbors=n, weights='distance')
#     model.fit(X_train, y_train)

#     predictions_train = model.predict(X_train)
#     F1_train = f1_score(y_train, predictions_train, average='macro')
#     scores_train.append(F1_train)

#     predictions = model.predict(X_test)
#     F1 = f1_score(y_test, predictions, average='macro')
#     scores.append(F1)

# # Plot the results
# plt.figure(figsize=(20, 8))
# plt.plot(neighbors_list, scores, lw=5)
# plt.plot(neighbors_list, scores_train, lw=5, color='orange')
# plt.xlabel('Number of neighbors')
# plt.ylabel('F1-Score')
# plt.legend(['F1: Test Set', 'F1: Train Set'])
# plt.show()
```

1.1.5 Decision Tree

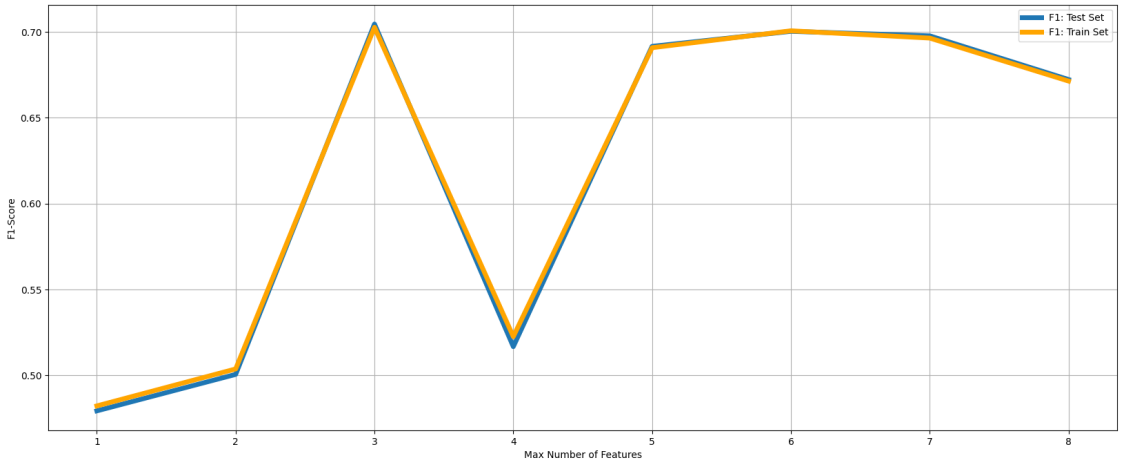
[46]:
```
def extract_feature_importance(decision_model,labels,decision_model_name='dt'):

  '''
  This function visualizes the important parameters
  '''

  labels_range = range(0,len(labels))
  if  decision_model_name == 'dt':
    trees = [decision_model]
  else:
    trees = decision_model.estimators_

  cnt = 1
  feature_importances = np.zeros([len(trees),len(labels)])
  itr = 0
```

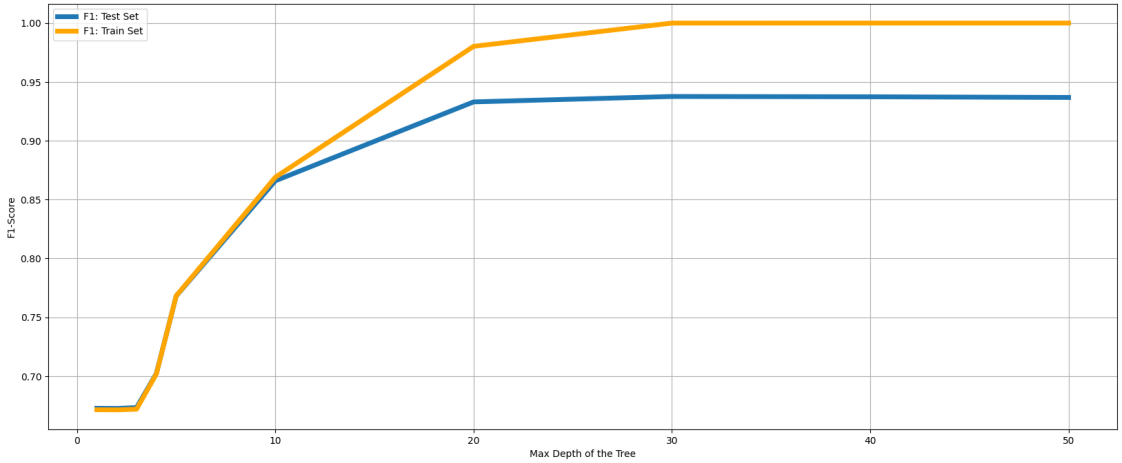
```
  for item in trees:
    feature_importances[itr,:] = item.feature_importances_
    itr += 1

  feature_importances_mean = feature_importances.mean(axis=0)
  feature_importances_std = feature_importances.std(axis=0)

  return feature_importances_mean
```

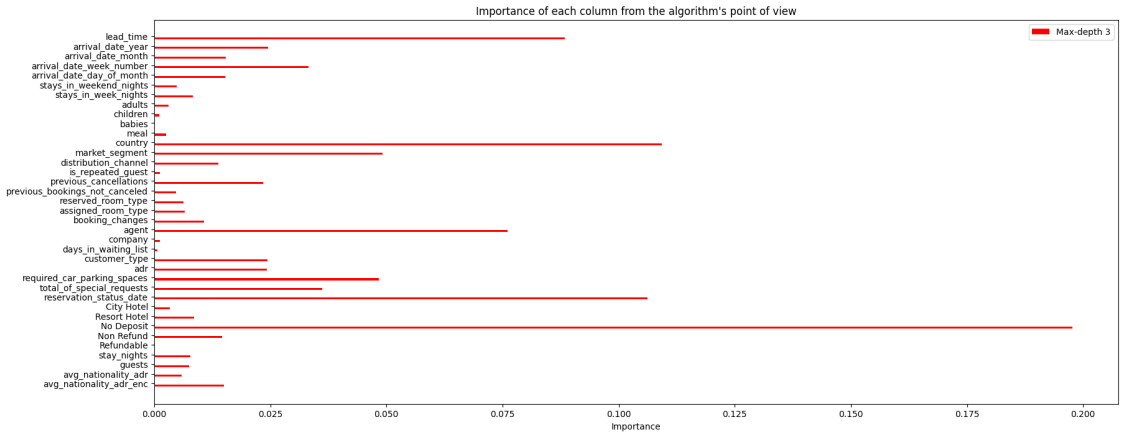
[69]:
```
# Prepering data
kkndf = odf.copy(True)
canceles = kkndf.pop("Canceled")
no_show = kkndf.pop("No-Show")
check_out = kkndf.pop("Check-Out")
X_train, X_test, y_train, y_test = train_test_split(kkndf, canceles,␣
 ↪test_size=0.3)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Searching for optimum number of features
scores = []
features_list = [1,2,3,4,5,6,7,8]
scores = []
scores_train = []

for no_features in features_list:
    model = DecisionTreeClassifier(max_features=no_features,max_depth=3)
    model.fit(X_train, y_train)
    predictions_train = model.predict(X_train)
    F1_train = f1_score(y_train, predictions_train, average='macro')
    scores_train.append(F1_train)

    predictions = model.predict(X_test)
    F1 = f1_score(y_test, predictions, average='macro')
    scores.append(F1)

plt.figure(figsize=(20,8))
plt.plot(features_list,scores,lw=5)
plt.plot(features_list,scores_train,lw=5,color='orange')
plt.xlabel('Max Number of Features')
plt.ylabel('F1-Score')
plt.legend(['F1: Test Set','F1: Train Set'])
plt.grid()
plt.show()

# Searching for optimum maximum depth
```

```
scores = []
depths_list = [1,2,3,4,5,10,20,30,40,50]
scores = []
scores_train = []
for depth in depths_list:
    model = DecisionTreeClassifier(max_depth=depth)
    model.fit(X_train, y_train)
    predictions_train = model.predict(X_train)
    F1_train = f1_score(y_train, predictions_train, average='macro')
    scores_train.append(F1_train)

    predictions = model.predict(X_test)
    F1 = f1_score(y_test, predictions, average='macro')
    scores.append(F1)

plt.figure(figsize=(20,8))
plt.plot(depths_list,scores,lw=5)
plt.plot(depths_list,scores_train,lw=5,color='orange')
plt.xlabel('Max Depth of the Tree')
plt.ylabel('F1-Score')
plt.legend(['F1: Test Set','F1: Train Set'])
plt.grid()
plt.show()

# Final model
model =␣
 ↪DecisionTreeClassifier(criterion='entropy',max_features=6,max_depth=20,min_samples_leaf=5)
model.fit(X_train, y_train)
predictions_train = model.predict(X_train)
predictions_test = model.predict(X_test)

# Visualizing features importance's
labels = list(knndf.columns)
features_importance =␣
 ↪extract_feature_importance(model,labels,decision_model_name='dt')
fig, ax = plt.subplots(figsize=(20,8))
y_pos = np.arange(0,len(features_importance),1)
hbars = ax.barh(y_pos+0.2, features_importance,␣
 ↪align='center',color='red',height=0.2)
ax.set_yticks(y_pos)
ax.set_yticklabels(labels)
ax.invert_yaxis()
ax.set_xlabel('Importance')
ax.set_title("Importance of each column from the algorithm's point of view")
plt.legend(['Max-depth 3','Max-depth 10'])
plt.show()
```

```python
# Plotting ROC curve
RocCurveDisplay.from_estimator(model, X_test, y_test)

# Metrics for train data
print("="*50)
print("TRAINING SET METRICS")
print("="*50)
print(f"Accuracy: {accuracy_score(y_train, predictions_train):.4f}")
print(f"Precision: {precision_score(y_train, predictions_train):.4f}")
print(f"Recall: {recall_score(y_train, predictions_train):.4f}")
print(f"F1-Score: {f1_score(y_train, predictions_train):.4f}")
print(f"AUC: {roc_auc_score(y_train, model.predict_proba(X_train)[:, 1]):.4f}")
print("\nConfusion Matrix:")
print(confusion_matrix(y_train, predictions_train))

# Metrics for test data
print("\n" + "="*50)
print("TEST SET METRICS")
print("="*50)
print(f"Accuracy: {accuracy_score(y_test, predictions_test):.4f}")
print(f"Precision: {precision_score(y_test, predictions_test):.4f}")
print(f"Recall: {recall_score(y_test, predictions_test):.4f}")
print(f"F1-Score: {f1_score(y_test, predictions_test):.4f}")
print(f"AUC: {roc_auc_score(y_test, model.predict_proba(X_test)[:, 1]):.4f}")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, predictions_test))
```

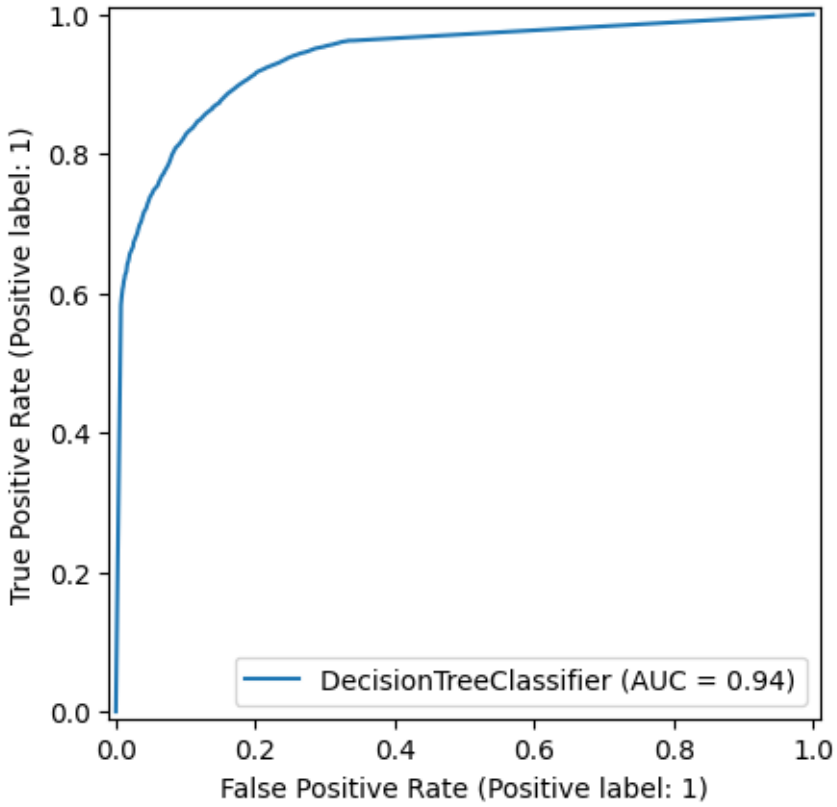







```
==================================================
TRAINING SET METRICS
==================================================
Accuracy: 0.9057
Precision: 0.8885
Recall: 0.8449
F1-Score: 0.8662
AUC: 0.9746

Confusion Matrix:
[[49969  3187]
 [ 4665 25407]]

==================================================
TEST SET METRICS
==================================================
```

```
Accuracy: 0.8763
Precision: 0.8430
Recall: 0.8077
F1-Score: 0.8250
AUC: 0.9375

Confusion Matrix:
[[20855  1937]
 [ 2476 10402]]
```



We could plot the tree and its leaves but the depth is too large and the number of leaves are too high that the tree chart does not contain any meaningful info.

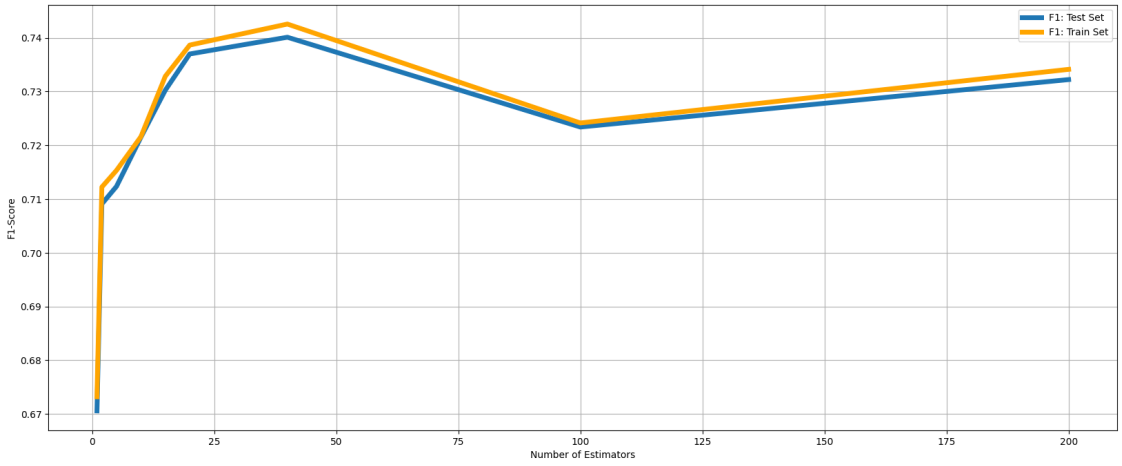
1.1.6 Random Forest

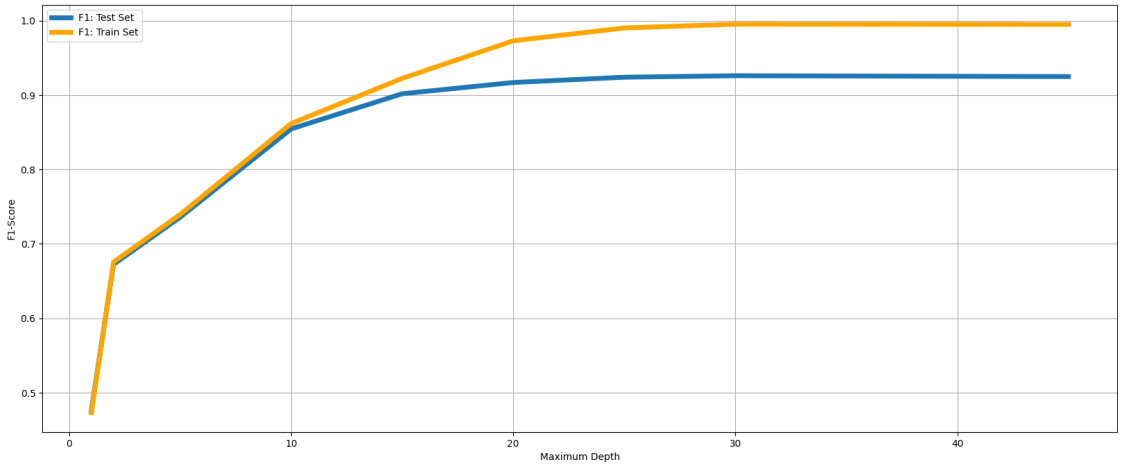
[ ]:
```
# Prepring data
rfdf = odf.copy(True)
canceles = rfdf.pop("Canceled")
no_show = rfdf.pop("No-Show")
check_out = rfdf.pop("Check-Out")
```

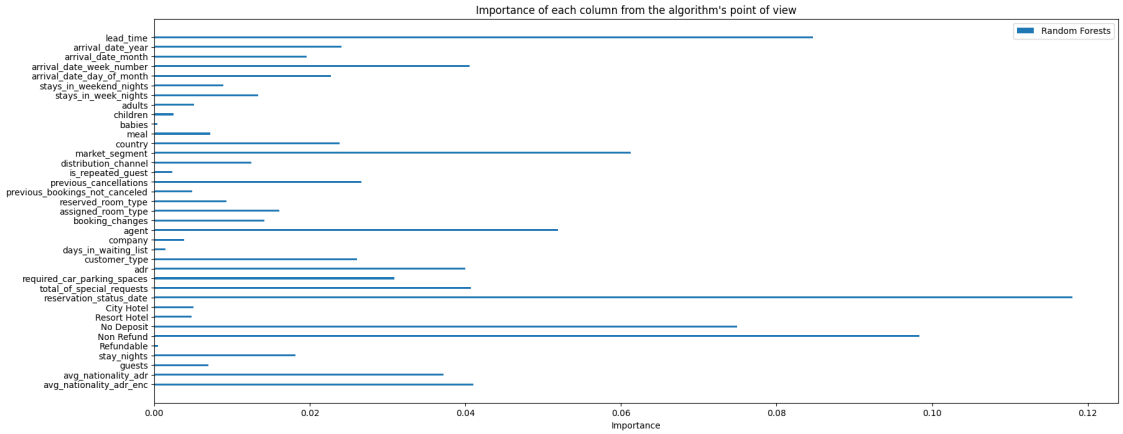
```
X_train, X_test, y_train, y_test = train_test_split(rfdf, canceles, test_size=0.
↪3)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Searching for optimum n_estimators
scores = []
no_estimators = [1,2,5,10,15,20,40,100,200]
scores = []
scores_train = []
for no_estimator in no_estimators:
    model = RandomForestClassifier(n_estimators=no_estimator,max_depth=5)
    model.fit(X_train, y_train)
    predictions_train = model.predict(X_train)
    F1_train = f1_score(y_train, predictions_train, average='macro')
    scores_train.append(F1_train)

    predictions = model.predict(X_test)
    F1 = f1_score(y_test, predictions, average='macro')
    scores.append(F1)

plt.figure(figsize=(20,8))
plt.plot(no_estimators,scores,lw=5)
plt.plot(no_estimators,scores_train,lw=5,color='orange')
plt.xlabel('Number of Estimators')
plt.ylabel('F1-Score')
plt.legend(['F1: Test Set','F1: Train Set'])
plt.grid()
plt.show()

# Searching for optimum max depth
max_depths = [1,2,5,10,15,20,25,30,45]
scores = []
scores_train = []
for md in max_depths:
    model = RandomForestClassifier(n_estimators=10,max_depth=md)
    model.fit(X_train, y_train)
    predictions_train = model.predict(X_train)
    F1_train = f1_score(y_train, predictions_train, average='macro')
    scores_train.append(F1_train)
    predictions = model.predict(X_test)
    F1 = f1_score(y_test, predictions, average='macro')
    scores.append(F1)

plt.figure(figsize=(20,8))
plt.plot(max_depths,scores,lw=5)
```

```
plt.plot(max_depths,scores_train,lw=5,color='orange')
plt.xlabel('Maximum Depth')
plt.ylabel('F1-Score')
plt.legend(['F1: Test Set','F1: Train Set'])
plt.grid()
plt.show()

# Final model
model =␣
 ↪RandomForestClassifier(n_estimators=100,criterion='entropy',max_depth=20)
model.fit(X_train, y_train)
predictions_train = model.predict(X_train)
predictions_test = model.predict(X_test)

# Visualizing feature importance
labels = list(rfdf.columns)
features_importance_rf =␣
 ↪extract_feature_importance(model,labels,decision_model_name='rf')
fig, ax = plt.subplots(figsize=(20,8))
y_pos = np.arange(0,len(features_importance_rf),1)
hbars = ax.barh(y_pos, features_importance_rf, align='center',height=.2)
ax.set_yticks(y_pos)
ax.set_yticklabels(labels)
ax.invert_yaxis()
ax.set_xlabel('Importance')
ax.set_title("Importance of each column from the algorithm's point of view")
plt.legend(['Random Forests','Decision Tree'])
plt.show()

# Metrics for train data
print("="*50)
print("TRAINING SET METRICS")
print("="*50)
print(f"Accuracy: {accuracy_score(y_train, predictions_train):.4f}")
print(f"Precision: {precision_score(y_train, predictions_train):.4f}")
print(f"Recall: {recall_score(y_train, predictions_train):.4f}")
print(f"F1-Score: {f1_score(y_train, predictions_train):.4f}")
print(f"AUC: {roc_auc_score(y_train, model.predict_proba(X_train)[:, 1]):.4f}")
print("\nConfusion Matrix:")
print(confusion_matrix(y_train, predictions_train))

#Metrics for test data
print("\n" + "="*50)
print("TEST SET METRICS")
print("="*50)
print(f"Accuracy: {accuracy_score(y_test, predictions_test):.4f}")
print(f"Precision: {precision_score(y_test, predictions_test):.4f}")
```

```
print(f"Recall: {recall_score(y_test, predictions_test):.4f}")
print(f"F1-Score: {f1_score(y_test, predictions_test):.4f}")
print(f"AUC: {roc_auc_score(y_test, model.predict_proba(X_test)[:, 1]):.4f}")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, predictions_test))
```








```
==================================================
TRAINING SET METRICS
==================================================
Accuracy: 0.9715
Precision: 0.9737
Recall: 0.9469
F1-Score: 0.9601
AUC: 0.9976

Confusion Matrix:
[[52310   772]
 [ 1601 28545]]

==================================================
TEST SET METRICS
==================================================
Accuracy: 0.9298
Precision: 0.9335
Recall: 0.8661
F1-Score: 0.8986
AUC: 0.9793

Confusion Matrix:
[[22076   790]
 [ 1714 11090]]
```

1.1.7 Multi Class Classification

[6]:
```
def multi_class_metrics(data,y_true, y_pred, class_names=None, y_proba=None):
    """
    Calculate multi-class classification metrics for each class
```

```
    Parameters:
    y_true: actual labels
    y_pred: predicted labels
    class_names: list of class names (optional)
    """

    # Get unique classes
    classes = np.unique(np.concatenate([y_true, y_pred]))
    n_classes = len(classes)

    if class_names is None:
        class_names = [f"Class_{c}" for c in classes]

    # Calculate confusion matrix
    cm = confusion_matrix(y_true, y_pred)

    # Initialize storage
    precision_per_class = []
    recall_per_class = []
    f1_per_class = []

    # Calculate metrics for each class (One-vs-All approach)
    for i, class_label in enumerate(classes):
        # True Positives: correctly predicted as this class
        tp = cm[i, i]

        # False Positives: predicted as this class but actual is other
        fp = cm[:, i].sum() - tp

        # False Negatives: actual is this class but predicted as other
        fn = cm[i, :].sum() - tp

        # True Negatives: actual and predicted are other classes
        tn = cm.sum() - tp - fp - fn

        # Calculate metrics for this class
        precision = tp / (tp + fp) if (tp + fp) > 0 else 0
        recall = tp / (tp + fn) if (tp + fn) > 0 else 0
        f1 = 2 * (precision * recall) / (precision + recall) if (precision +␣
↪recall) > 0 else 0

        precision_per_class.append(precision)
        recall_per_class.append(recall)
        f1_per_class.append(f1)

    # Calculate macro, micro, weighted averages
    macro_precision = np.mean(precision_per_class)
```

```
    macro_recall = np.mean(recall_per_class)
    macro_f1 = np.mean(f1_per_class)

    # Micro: aggregate totals first
    total_tp = sum([cm[i, i] for i in range(n_classes)])
    total_fp = sum([cm[:, i].sum() - cm[i, i] for i in range(n_classes)])
    total_fn = sum([cm[i, :].sum() - cm[i, i] for i in range(n_classes)])

    micro_precision = total_tp / (total_tp + total_fp) if (total_tp + total_fp)␣
↪> 0 else 0
    micro_recall = total_tp / (total_tp + total_fn) if (total_tp + total_fn) >␣
↪0 else 0
    micro_f1 = 2 * (micro_precision * micro_recall) / (micro_precision +␣
↪micro_recall) if (micro_precision + micro_recall) > 0 else 0

    # Weighted: weighted by support (number of true instances per class)
    support = [cm[i, :].sum() for i in range(n_classes)]
    total_samples = sum(support)

    weighted_precision = sum([p * s for p, s in zip(precision_per_class,␣
↪support)]) / total_samples
    weighted_recall = sum([r * s for r, s in zip(recall_per_class, support)]) /␣
↪total_samples
    weighted_f1 = sum([f * s for f, s in zip(f1_per_class, support)]) /␣
↪total_samples

    kappa = cohen_kappa_score(y_true, y_pred)

    # Create results DataFrame
    results_df = pd.DataFrame({
        'Class': class_names,
        'Precision': precision_per_class,
        'Recall': recall_per_class,
        'F1-Score': f1_per_class,
        'Support': support
    })

    # Display results
    print("="*70)
    print(f"MULTI-CLASS CLASSIFICATION METRICS ({data})")
    print("="*70)
    print(results_df.to_string(index=False, float_format='%.4f'))

    print("\n" + "="*70)
    print(f"AGGREGATED METRICS ({data})")
    print("="*70)
```

```
    print(f"\nAccuracy: {accuracy_score(y_true, y_pred):.4f}")
    print(f"\nMacro Average:")
    print(f"  Precision: {macro_precision:.4f}")
    print(f"  Recall:    {macro_recall:.4f}")
    print(f"  F1-Score:  {macro_f1:.4f}")
    print(f"\nMicro Average:")
    print(f"  Precision: {micro_precision:.4f}")
    print(f"  Recall:    {micro_recall:.4f}")
    print(f"  F1-Score:  {micro_f1:.4f}")
    print(f"\nWeighted Average:")
    print(f"  Precision: {weighted_precision:.4f}")
    print(f"  Recall:    {weighted_recall:.4f}")
    print(f"  F1-Score:  {weighted_f1:.4f}")
    print(f"\nOverall Kappa: {kappa:.4f}")  # ONE number

    if y_proba is not None:
        ll = log_loss(y_true,y_proba)
        print(f"\nLog Loss: {ll:.4f}")

    # Confusion matrix shows per-class performance
    cm = confusion_matrix(y_true, y_pred)
    print("\nConfusion Matrix:")
    print(cm)
    return
```

[56]:
```
mcdf = odf.copy(True)
numerizing_kernel = {'Canceled': 0, 'No-Show': 1, 'Check-Out': 2}
target = mcdf[list(numerizing_kernel.keys())].dot(list(numerizing_kernel.
 ↪values()))
canceles = mcdf.pop("Canceled")
no_shows = mcdf.pop("No-Show")
check_out = mcdf.pop("Check-Out")

X_train, X_test, y_train, y_test = train_test_split(mcdf, target, test_size=0.3)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.fit_transform(X_test)
```

1.1.8 SVM Linear Kernel

[57]:
```
model = SVC(C=3,kernel='linear',decision_function_shape =␣
 ↪"ovr",max_iter=5000,class_weight={0: 1, 1: 28, 2: 1})
model.fit(X_train, y_train)
predictions_train = model.predict(X_train)
predictions_test = model.predict(X_test)
```

```
multi_class_metrics('Trainning',y_train,predictions_train,class_names=numerizing_kernel.
  ↪keys())
multi_class_metrics('Testing',y_test,predictions_test,class_names=numerizing_kernel.
  ↪keys())
```

```
/Users/parsajoneydi/.pyenv/versions/scientific/lib/python3.14/site-
packages/sklearn/svm/_base.py:313: ConvergenceWarning: Solver terminated early
(max_iter=5000).  Consider pre-processing your data with StandardScaler or
MinMaxScaler.
  warnings.warn(

======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Trainning)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.6029  0.8445    0.7036    30151
  No-Show     0.0128  0.1163    0.0230      808
Check-Out     0.8958  0.5764    0.7014    52269

======================================================================
AGGREGATED METRICS (Trainning)
======================================================================

Accuracy: 0.6691

Macro Average:
  Precision: 0.5038
  Recall:    0.5124
  F1-Score:  0.4760

Micro Average:
  Precision: 0.6691
  Recall:    0.6691
  F1-Score:  0.6691

Weighted Average:
  Precision: 0.7811
  Recall:    0.6691
  F1-Score:  0.6956

Overall Kappa: 0.4106

Confusion Matrix:
[[25464  1652  3035]
 [  243    94   471]
 [16528  5614 30127]]
======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Testing)
```

```
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.5992  0.8458    0.7014    12799
  No-Show     0.0144  0.1190    0.0257      395
Check-Out     0.8978  0.5727    0.6994    22476

======================================================================
AGGREGATED METRICS (Testing)
======================================================================

Accuracy: 0.6657

Macro Average:
  Precision: 0.5038
  Recall:    0.5125
  F1-Score:  0.4755

Micro Average:
  Precision: 0.6657
  Recall:    0.6657
  F1-Score:  0.6657

Weighted Average:
  Precision: 0.7809
  Recall:    0.6657
  F1-Score:  0.6926

Overall Kappa: 0.4072

Confusion Matrix:
[[10825   737  1237]
 [  120    47   228]
 [ 7121  2482 12873]]
```

1.1.9 SVM RBF Kernel

[ ]:
```
model = SVC(C=2,kernel='rbf',decision_function_shape =␣
 ↪"ovo",max_iter=5000,class_weight={0: 1, 1: 28, 2: 1})
model.fit(X_train, y_train)
predictions_train = model.predict(X_train)
predictions_test = model.predict(X_test)

multi_class_metrics('Trainning',y_train,predictions_train,class_names=numerizing_kernel.
 ↪keys())
multi_class_metrics('Testing',y_test,predictions_test,class_names=numerizing_kernel.
 ↪keys())
```

```
/Users/parsajoneydi/.pyenv/versions/scientific/lib/python3.14/site-
```

```
packages/sklearn/svm/_base.py:313: ConvergenceWarning: Solver terminated early
(max_iter=5000).  Consider pre-processing your data with StandardScaler or
MinMaxScaler.
  warnings.warn(

======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Trainning)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9907  0.5183    0.6806    52181
  No-Show     0.9529  0.4448    0.6065    52181
Check-Out     0.4903  0.9856    0.6549    52181

======================================================================
AGGREGATED METRICS (Trainning)
======================================================================

Accuracy: 0.6496

Macro Average:
  Precision: 0.8113
  Recall:    0.6496
  F1-Score:  0.6473

Micro Average:
  Precision: 0.6496
  Recall:    0.6496
  F1-Score:  0.6496

Weighted Average:
  Precision: 0.8113
  Recall:    0.6496
  F1-Score:  0.6473

Overall Kappa: 0.4744

Confusion Matrix:
[[27047   525 24609]
 [  124 23208 28849]
 [  129   621 51431]]
======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Testing)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9485  0.5103    0.6636    22564
  No-Show     0.8971  0.2161    0.3482    22564
Check-Out     0.4426  0.9832    0.6104    22564
```

```
======================================================================
AGGREGATED METRICS (Testing)
======================================================================

Accuracy: 0.5698

Macro Average:
  Precision: 0.7628
  Recall:    0.5698
  F1-Score:  0.5407

Micro Average:
  Precision: 0.5698
  Recall:    0.5698
  F1-Score:  0.5698

Weighted Average:
  Precision: 0.7628
  Recall:    0.5698
  F1-Score:  0.5407

Overall Kappa: 0.3547

Confusion Matrix:
[[11514   245 10805]
 [  559  4875 17130]
 [   66   314 22184]]
```

1.1.10 Logistic Regression (OVR)

[58]:
```
model =␣
 ↪LogisticRegression(penalty='l2',max_iter=5000,solver='newton-cg',class_weight={0:
 ↪ 1, 1: 28, 2: 1})
model.fit(X_train, y_train)
predictions_train = model.predict(X_train)
predictions_test = model.predict(X_test)
predictions_train_proba = model.predict_proba(X_train)
predictions_test_proba = model.predict_proba(X_test)

multi_class_metrics('Trainning',y_train,predictions_train,class_names=numerizing_kernel.
 ↪keys(),y_proba=predictions_train_proba)
multi_class_metrics('Testing',y_test,predictions_test,class_names=numerizing_kernel.
 ↪keys(),y_proba=predictions_test_proba)
```

```
/Users/parsajoneydi/.pyenv/versions/scientific/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
```

```
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
  warnings.warn(

======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Trainning)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9973  0.8919    0.9417    30151
  No-Show     0.0691  0.5111    0.1217      808
Check-Out     0.9659  0.9292    0.9472    52269

======================================================================
AGGREGATED METRICS (Trainning)
======================================================================

Accuracy: 0.9116

Macro Average:
  Precision: 0.6774
  Recall:    0.7774
  F1-Score:  0.6702

Micro Average:
  Precision: 0.9116
  Recall:    0.9116
  F1-Score:  0.9116

Weighted Average:
  Precision: 0.9685
  Recall:    0.9116
  F1-Score:  0.9372

Overall Kappa: 0.8242

Log Loss: 0.2807

Confusion Matrix:
[[26893  1936  1322]
 [    2   413   393]
 [   72  3629 48568]]
======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Testing)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9959  0.8926    0.9415    12799
  No-Show     0.0765  0.4734    0.1318      395
Check-Out     0.9650  0.9340    0.9492    22476
```

```
======================================================================
AGGREGATED METRICS (Testing)
======================================================================

Accuracy: 0.9141

Macro Average:
  Precision: 0.6791
  Recall:    0.7667
  F1-Score:  0.6742

Micro Average:
  Precision: 0.9141
  Recall:    0.9141
  F1-Score:  0.9141

Weighted Average:
  Precision: 0.9662
  Recall:    0.9141
  F1-Score:  0.9374

Overall Kappa: 0.8280

Log Loss: 0.2841

Confusion Matrix:
[[11425   817   557]
 [    3   187   205]
 [   44  1439 20993]]
```

[60]:
```
model =␣
 ↪LogisticRegression(penalty='l2',max_iter=5000,solver='lbfgs',class_weight={0:
 ↪ 1, 1: 28, 2: 1})
model.fit(X_train, y_train)
predictions_train = model.predict(X_train)
predictions_test = model.predict(X_test)
predictions_train_proba = model.predict_proba(X_train)
predictions_test_proba = model.predict_proba(X_test)

multi_class_metrics('Trainning',y_train,predictions_train,class_names=numerizing_kernel.
 ↪keys(),y_proba=predictions_train_proba)
multi_class_metrics('Testing',y_test,predictions_test,class_names=numerizing_kernel.
 ↪keys(),y_proba=predictions_test_proba)
```

```
/Users/parsajoneydi/.pyenv/versions/scientific/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
```

```
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
  warnings.warn(

======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Trainning)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9972  0.8930    0.9422    30151
  No-Show     0.0694  0.5136    0.1223      808
Check-Out     0.9662  0.9288    0.9471    52269

======================================================================
AGGREGATED METRICS (Trainning)
======================================================================

Accuracy: 0.9118

Macro Average:
  Precision: 0.6776
  Recall:    0.7785
  F1-Score:  0.6705

Micro Average:
  Precision: 0.9118
  Recall:    0.9118
  F1-Score:  0.9118

Weighted Average:
  Precision: 0.9687
  Recall:    0.9118
  F1-Score:  0.9373

Overall Kappa: 0.8245

Log Loss: 0.2791

Confusion Matrix:
[[26924  1917  1310]
 [    3   415   390]
 [   73  3648 48548]]
======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Testing)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9959  0.8945    0.9425    12799
  No-Show     0.0763  0.4709    0.1313      395
```

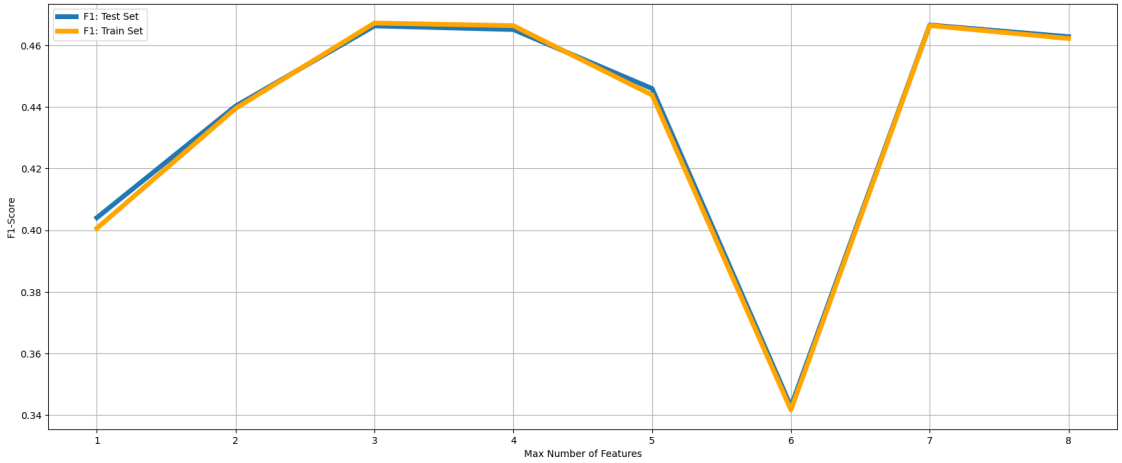
```
Check-Out      0.9653   0.9334     0.9491     22476

================================================================================
AGGREGATED METRICS (Testing)
================================================================================

Accuracy: 0.9144

Macro Average:
  Precision: 0.6791
  Recall:    0.7663
  F1-Score:  0.6743

Micro Average:
  Precision: 0.9144
  Recall:    0.9144
  F1-Score:  0.9144

Weighted Average:
  Precision: 0.9664
  Recall:    0.9144
  F1-Score:  0.9377

Overall Kappa: 0.8286

Log Loss: 0.2830

Confusion Matrix:
[[11449   801   549]
 [    3   186   206]
 [   44  1452 20980]]
```

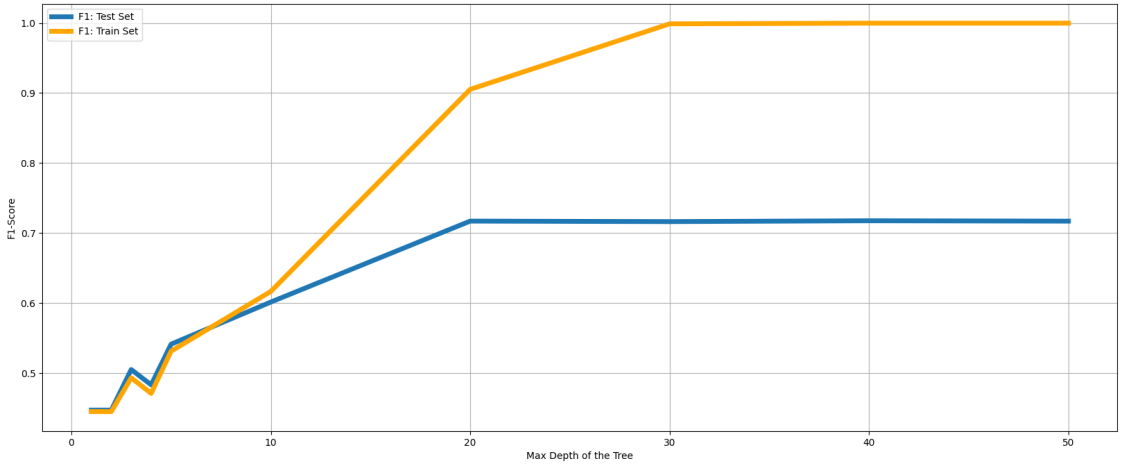
1.1.11 Multi Class Decision Tree

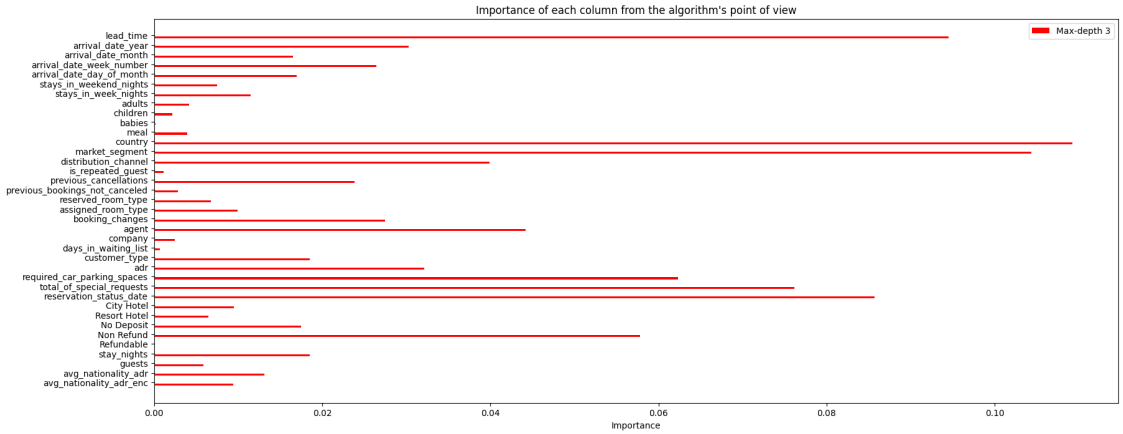
[61]:
```
# Searching for optimum number of features
features_list = [1,2,3,4,5,6,7,8]
scores = []
scores_train = []
for no_features in features_list:
    model = DecisionTreeClassifier(max_features=no_features,max_depth=3)
    model.fit(X_train, y_train)
    predictions_train = model.predict(X_train)
    F1_train = f1_score(y_train, predictions_train, average='macro')
    scores_train.append(F1_train)
    predictions = model.predict(X_test)
    F1 = f1_score(y_test, predictions, average='macro')
    scores.append(F1)
```

```
plt.figure(figsize=(20,8))
plt.plot(features_list,scores,lw=5)
plt.plot(features_list,scores_train,lw=5,color='orange')
plt.xlabel('Max Number of Features')
plt.ylabel('F1-Score')
plt.legend(['F1: Test Set','F1: Train Set'])
plt.grid()
plt.show()

# Searching for optimum maximum depth
depths_list = [1,2,3,4,5,10,20,30,40,50]
scores = []
scores_train = []
for depth in depths_list:
    model = DecisionTreeClassifier(max_depth=depth)
    model.fit(X_train, y_train)
    predictions_train = model.predict(X_train)
    F1_train = f1_score(y_train, predictions_train, average='macro')
    scores_train.append(F1_train)
    predictions = model.predict(X_test)
    F1 = f1_score(y_test, predictions, average='macro')
    scores.append(F1)
plt.figure(figsize=(20,8))
plt.plot(depths_list,scores,lw=5)
plt.plot(depths_list,scores_train,lw=5,color='orange')
plt.xlabel('Max Depth of the Tree')
plt.ylabel('F1-Score')
plt.legend(['F1: Test Set','F1: Train Set'])
plt.grid()
plt.show()

# Final model
model =␣
 ↪DecisionTreeClassifier(criterion='entropy',max_features=6,max_depth=20,min_samples_leaf=5)
model.fit(X_train, y_train)
predictions_train = model.predict(X_train)
predictions_test = model.predict(X_test)
predictions_train_porba = model.predict_proba(X_train)
predictions_test_porba = model.predict_proba(X_test)

# Visualizing features importance's
labels = list(mcdf.columns)
features_importance =␣
 ↪extract_feature_importance(model,labels,decision_model_name='dt')
fig, ax = plt.subplots(figsize=(20,8))
y_pos = np.arange(0,len(features_importance),1)
```

```
hbars = ax.barh(y_pos+0.2, features_importance,␣
 ↪align='center',color='red',height=0.2)
ax.set_yticks(y_pos)
ax.set_yticklabels(labels)
ax.invert_yaxis()
ax.set_xlabel('Importance')
ax.set_title("Importance of each column from the algorithm's point of view")
plt.legend(['Max-depth 3','Max-depth 10'])
plt.show()

# Print Metrics
multi_class_metrics('Trainning',y_train,predictions_train,class_names=numerizing_kernel.
 ↪keys(),y_proba=predictions_train_proba)
multi_class_metrics('Testing',y_test,predictions_test,class_names=numerizing_kernel.
 ↪keys(),y_proba=predictions_test_proba)
```








```
========================================================================
MULTI-CLASS CLASSIFICATION METRICS (Trainning)
========================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.8830  0.8427    0.8624    30151
  No-Show     0.6246  0.2389    0.3456      808
Check-Out     0.9053  0.9378    0.9212    52269

========================================================================
AGGREGATED METRICS (Trainning)
========================================================================

Accuracy: 0.8965

Macro Average:
  Precision: 0.8043
  Recall:    0.6731
  F1-Score:  0.7097

Micro Average:
  Precision: 0.8965
  Recall:    0.8965
  F1-Score:  0.8965

Weighted Average:
  Precision: 0.8945
  Recall:    0.8965
  F1-Score:  0.8943

Overall Kappa: 0.7780

Log Loss: 0.2791
```

```
Confusion Matrix:
[[25407    49  4695]
 [  180   193   435]
 [ 3186    67 49016]]
======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Testing)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.8259  0.7892    0.8071    12799
  No-Show     0.2321  0.0658    0.1026      395
Check-Out     0.8740  0.9071    0.8902    22476

======================================================================
AGGREGATED METRICS (Testing)
======================================================================

Accuracy: 0.8555

Macro Average:
  Precision: 0.6440
  Recall:    0.5874
  F1-Score:  0.6000

Micro Average:
  Precision: 0.8555
  Recall:    0.8555
  F1-Score:  0.8555

Weighted Average:
  Precision: 0.8496
  Recall:    0.8555
  F1-Score:  0.8517

Overall Kappa: 0.6891

Log Loss: 0.2830

Confusion Matrix:
[[10101    30  2668]
 [   97    26   272]
 [ 2033    56 20387]]
```

1.1.12 XGBoost

[62]:
```
model = XGBClassifier(n_estimators=50,eval_metric='logloss',objective="multi:
 ↪softmax")
model.fit(X_train, y_train)
predictions_train = model.predict(X_train)
predictions_test = model.predict(X_test)
predictions_train_proba = model.predict_proba(X_train)
predictions_test_proba = model.predict_proba(X_test)

multi_class_metrics('Trainning',y_train,predictions_train,class_names=numerizing_kernel.
 ↪keys(),y_proba=predictions_train_proba)
multi_class_metrics('Testing',y_test,predictions_test,class_names=numerizing_kernel.
 ↪keys(),y_proba=predictions_test_proba)
```

```
======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Trainning)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9926  0.9268    0.9586    30151
  No-Show     0.9524  0.2475    0.3929      808
Check-Out     0.9496  0.9968    0.9727    52269

======================================================================
AGGREGATED METRICS (Trainning)
======================================================================

Accuracy: 0.9642

Macro Average:
  Precision: 0.9649
  Recall:    0.7237
  F1-Score:  0.7747

Micro Average:
  Precision: 0.9642
  Recall:    0.9642
  F1-Score:  0.9642

Weighted Average:
  Precision: 0.9652
  Recall:    0.9642
  F1-Score:  0.9619

Overall Kappa: 0.9227

Log Loss: 0.1301
```

```
Confusion Matrix:
[[27944     6  2201]
 [   46   200   562]
 [  162     4 52103]]
======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Testing)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9889  0.8102    0.8907    12799
  No-Show     0.5714  0.0405    0.0757      395
Check-Out     0.8893  0.9953    0.9393    22476

======================================================================
AGGREGATED METRICS (Testing)
======================================================================

Accuracy: 0.9183

Macro Average:
  Precision: 0.8165
  Recall:    0.6153
  F1-Score:  0.6352

Micro Average:
  Precision: 0.9183
  Recall:    0.9183
  F1-Score:  0.9183

Weighted Average:
  Precision: 0.9215
  Recall:    0.9183
  F1-Score:  0.9123

Overall Kappa: 0.8185

Log Loss: 0.2416

Confusion Matrix:
[[10370     9  2420]
 [   13    16   366]
 [  103     3 22370]]
```

1.1.13 LightGBM

[63]:
```
model = lgbm.LGBMClassifier(learning_rate=0.09,max_depth=10)
model.fit(X_train,y_train,eval_metric=log_loss)
predictions_train = model.predict(X_train)
```

```python
predictions_test = model.predict(X_test)
predictions_train_proba = model.predict_proba(X_train)
predictions_test_proba = model.predict_proba(X_test)

multi_class_metrics('Trainning',y_train,predictions_train,class_names=numerizing_kernel.
 ↪keys(),y_proba=predictions_train_proba)
multi_class_metrics('Testing',y_test,predictions_test,class_names=numerizing_kernel.
 ↪keys(),y_proba=predictions_test_proba)
```

```
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.006798 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 2051
[LightGBM] [Info] Number of data points in the train set: 83228, number of used
features: 37
[LightGBM] [Info] Start training from score -1.015366
[LightGBM] [Info] Start training from score -4.634777
[LightGBM] [Info] Start training from score -0.465180
```

```
/Users/parsajoneydi/.pyenv/versions/scientific/lib/python3.14/site-
packages/sklearn/utils/validation.py:2691: UserWarning: X does not have valid
feature names, but LGBMClassifier was fitted with feature names
  warnings.warn(
/Users/parsajoneydi/.pyenv/versions/scientific/lib/python3.14/site-
packages/sklearn/utils/validation.py:2691: UserWarning: X does not have valid
feature names, but LGBMClassifier was fitted with feature names
  warnings.warn(
/Users/parsajoneydi/.pyenv/versions/scientific/lib/python3.14/site-
packages/sklearn/utils/validation.py:2691: UserWarning: X does not have valid
feature names, but LGBMClassifier was fitted with feature names
  warnings.warn(
/Users/parsajoneydi/.pyenv/versions/scientific/lib/python3.14/site-
packages/sklearn/utils/validation.py:2691: UserWarning: X does not have valid
feature names, but LGBMClassifier was fitted with feature names
  warnings.warn(
```

```
======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Trainning)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9903  0.9251    0.9566    30151
  No-Show     0.9567  0.3007    0.4576      808
Check-Out     0.9494  0.9955    0.9719    52269

======================================================================
AGGREGATED METRICS (Trainning)
======================================================================
```

```
Accuracy: 0.9633

Macro Average:
  Precision: 0.9655
  Recall:    0.7405
  F1-Score:  0.7954

Micro Average:
  Precision: 0.9633
  Recall:    0.9633
  F1-Score:  0.9633

Weighted Average:
  Precision: 0.9643
  Recall:    0.9633
  F1-Score:  0.9614

Overall Kappa: 0.9208

Log Loss: 0.1248

Confusion Matrix:
[[27893     7  2251]
 [   43   243   522]
 [  230     4 52035]]
======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Testing)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9882  0.9169    0.9512    12799
  No-Show     0.7797  0.1165    0.2026      395
Check-Out     0.9419  0.9947    0.9676    22476

======================================================================
AGGREGATED METRICS (Testing)
======================================================================

Accuracy: 0.9571

Macro Average:
  Precision: 0.9033
  Recall:    0.6760
  F1-Score:  0.7072

Micro Average:
  Precision: 0.9571
  Recall:    0.9571
  F1-Score:  0.9571
```

```
Weighted Average:
  Precision: 0.9567
  Recall:    0.9571
  F1-Score:  0.9532

Overall Kappa: 0.9069

Log Loss: 0.1520

Confusion Matrix:
[[11736     6  1057]
 [   27    46   322]
 [  113     7 22356]]
```

1.1.14 Adaboost

[71]:
```
base_model = DecisionTreeClassifier(criterion='log_loss',
    max_depth=10,
    max_features=5,
    min_samples_split=10,
    min_samples_leaf=5
)
model =␣
 ↪AdaBoostClassifier(estimator=base_model,n_estimators=100,learning_rate=0.5,)
model.fit(X_train, y_train)
predictions_train = model.predict(X_train)
predictions_test = model.predict(X_test)
predictions_train_proba = model.predict_proba(X_train)
predictions_test_proba = model.predict_proba(X_test)

multi_class_metrics('Trainning',y_train,predictions_train,class_names=numerizing_kernel.
 ↪keys(),y_proba=predictions_train_proba)
multi_class_metrics('Testing',y_test,predictions_test,class_names=numerizing_kernel.
 ↪keys(),y_proba=predictions_test_proba)
```

```
======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Trainning)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9637  0.9330    0.9481    30151
  No-Show     0.9867  0.6423    0.7781      808
Check-Out     0.9578  0.9806    0.9691    52269

======================================================================
AGGREGATED METRICS (Trainning)
======================================================================
```

```
Accuracy: 0.9601

Macro Average:
  Precision: 0.9694
  Recall:    0.8520
  F1-Score:  0.8984

Micro Average:
  Precision: 0.9601
  Recall:    0.9601
  F1-Score:  0.9601

Weighted Average:
  Precision: 0.9603
  Recall:    0.9601
  F1-Score:  0.9596

Overall Kappa: 0.9149

Log Loss: 0.9368

Confusion Matrix:
[[28130     3  2018]
 [   51   519   238]
 [ 1008     4 51257]]
======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Testing)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9241  0.8727    0.8977    12799
  No-Show     0.8800  0.1671    0.2809      395
Check-Out     0.9200  0.9622    0.9407    22476

======================================================================
AGGREGATED METRICS (Testing)
======================================================================

Accuracy: 0.9213

Macro Average:
  Precision: 0.9080
  Recall:    0.6673
  F1-Score:  0.7064

Micro Average:
  Precision: 0.9213
  Recall:    0.9213
  F1-Score:  0.9213
```

```
Weighted Average:
  Precision: 0.9210
  Recall:    0.9213
  F1-Score:  0.9179

Overall Kappa: 0.8301

Log Loss: 0.9422

Confusion Matrix:
[[11170     4  1625]
 [   74    66   255]
 [  844     5 21627]]
```

1.1.15 Cat Boost

[85]:
```
# Basic CatBoost model (handles categorical features automatically
model = CatBoostClassifier(
    iterations=100,
    learning_rate=0.1,
    depth=6,
    loss_function='MultiClass',
    eval_metric='Accuracy',
    verbose=50
)

# Train
model.fit(X_train, y_train)

# Predict
predictions_train = model.predict(X_train).flatten()
predictions_test = model.predict(X_test).flatten()
predictions_train_proba = model.predict_proba(X_train)
predictions_test_proba = model.predict_proba(X_test)

multi_class_metrics('Training', y_train, predictions_train,
                    class_names=numerizing_kernel.keys(),
                    y_proba=predictions_train_proba)
multi_class_metrics('Testing', y_test, predictions_test,
                    class_names=numerizing_kernel.keys(),
                    y_proba=predictions_test_proba)
```

```
0:      learn: 0.7934349        total: 18ms     remaining: 1.78s
50:     learn: 0.8782261        total: 819ms    remaining: 787ms
99:     learn: 0.9094896        total: 1.66s    remaining: 0us
=====================================================================
MULTI-CLASS CLASSIFICATION METRICS (Training)
```

```
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9361  0.8278    0.8786    30151
  No-Show     0.8800  0.0272    0.0528      808
Check-Out     0.8970  0.9702    0.9322    52269

======================================================================
AGGREGATED METRICS (Training)
======================================================================

Accuracy: 0.9095

Macro Average:
  Precision: 0.9043
  Recall:    0.6084
  F1-Score:  0.6212

Micro Average:
  Precision: 0.9095
  Recall:    0.9095
  F1-Score:  0.9095

Weighted Average:
  Precision: 0.9110
  Recall:    0.9095
  F1-Score:  0.9042

Overall Kappa: 0.8021

Log Loss: 0.2659

Confusion Matrix:
[[24960     2  5189]
 [  150    22   636]
 [ 1555     1 50713]]
======================================================================
MULTI-CLASS CLASSIFICATION METRICS (Testing)
======================================================================
    Class  Precision  Recall  F1-Score  Support
 Canceled     0.9288  0.8224    0.8724    12799
  No-Show     0.6667  0.0101    0.0200      395
Check-Out     0.8939  0.9677    0.9293    22476

======================================================================
AGGREGATED METRICS (Testing)
======================================================================

Accuracy: 0.9049
```

```
Macro Average:
  Precision: 0.8298
  Recall:    0.6001
  F1-Score:  0.6072

Micro Average:
  Precision: 0.9049
  Recall:    0.9049
  F1-Score:  0.9049

Weighted Average:
  Precision: 0.9039
  Recall:    0.9049
  F1-Score:  0.8988

Overall Kappa: 0.7916

Log Loss: 0.2742

Confusion Matrix:
[[10526     1  2272]
 [   81     4   310]
 [  726     1 21749]]
```

Instead of giving weight to classes to compensate for the lack of no-show data rows, we could have resampled the data; which was done but resulted in lots of overfitting and messy confusion matrices so much so that in the end it is preferable to stay with weighted classes.

1.1.16 Comparison (Binary Class vs Multi Class)

We use LightGBM as the repsentative of Multi-Class classification and Random Forest as the represntative of binray classification methods for comparison of the two methods on this problem.

[ ]:
```python
# Preperring data
df = odf.copy(True)

# Create target variable with 3 classes
def create_multiclass_target(row):
    if row['Canceled'] == 1:
        return 'Canceled'
    elif row['No-Show'] == 1:
        return 'No-Show'
    else:
        return 'Check-Out'
```

```
df['Reservation_Status'] = df.apply(create_multiclass_target, axis=1)

# Remove original binary columns
df = df.drop(['Canceled', 'No-Show', 'Check-Out'], axis=1)

# Encode target
label_encoder = LabelEncoder()
y = label_encoder.fit_transform(df['Reservation_Status'])
class_names = label_encoder.classes_  # ['Canceled', 'Check-Out', 'No-Show']

# Remove target from features
X = rfdf.drop('Reservation_Status', axis=1)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y
)

# Scale features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print(f"Training set size: {X_train.shape[0]}")
print(f"Test set size: {X_test.shape[0]}")
print(f"Classes: {class_names}")
print(f"Class distribution:\n{pd.Series(y_train).value_counts()}")

# # Random Forest (Binray)
# print("\n" + "="*60)
# print("RANDOM FOREST - BINARYCLASS")
# print("="*60)

# # Hyperparameter tuning for RF
# rf_params = {
#     'n_estimators': [50, 100, 200],
#     'max_depth': [10, 15, 20, None],
#     'min_samples_split': [5, 10, 15],
#     'min_samples_leaf': [2, 4, 6]
# }

# # Best parameters from your analysis
# best_rf = RandomForestClassifier(
#     n_estimators=100,
#     max_depth=20,
#     min_samples_split=10,
#     min_samples_leaf=4,
```

```
#     criterion='gini',  # 'gini' often better for multiclass
#     random_state=42,
#     n_jobs=-1
# )

# best_rf.fit(X_train_scaled, y_train)

# # Predictions
# rf_train_pred = best_rf.predict(X_train_scaled)
# rf_test_pred = best_rf.predict(X_test_scaled)
# rf_train_proba = best_rf.predict_proba(X_train_scaled)
# rf_test_proba = best_rf.predict_proba(X_test_scaled)

# # Metrics for RF
# rf_train_acc = accuracy_score(y_train, rf_train_pred)
# rf_test_acc = accuracy_score(y_test, rf_test_pred)
# rf_train_f1 = f1_score(y_train, rf_train_pred, average='macro')
# rf_test_f1 = f1_score(y_test, rf_test_pred, average='macro')

# print(f"\nRandom Forest Performance:")
# print(f"  Train Accuracy: {rf_train_acc:.4f}")
# print(f"  Test Accuracy:  {rf_test_acc:.4f}")
# print(f"  Overfitting Gap: {rf_train_acc - rf_test_acc:.4f}")
# print(f"  Train F1 (macro): {rf_train_f1:.4f}")
# print(f"  Test F1 (macro):  {rf_test_f1:.4f}")

print("\n" + "="*60)
print("RANDOM FOREST - BINARY CLASSIFIERS WITH VOTING")
print("="*60)

class BinaryRandomForestWithVoting:
    def __init__(self, n_estimators=100, max_depth=20, min_samples_split=10,
                 min_samples_leaf=4, random_state=42):
        """
        Initialize binary classifiers for each class
        """
        self.class_names = None
        self.classifiers = {}
        self.params = {
            'n_estimators': n_estimators,
            'max_depth': max_depth,
            'min_samples_split': min_samples_split,
            'min_samples_leaf': min_samples_leaf,
            'criterion': 'gini',
            'random_state': random_state,
            'n_jobs': -1
        }
```

```
    def fit(self, X, y, class_names):
        """
        Train one binary Random Forest per class
        """
        self.class_names = class_names

        for class_name in class_names:
            print(f"\nTraining binary classifier for: {class_name}")

            # Create binary labels (1 for current class, 0 for others)
            y_binary = (y == class_name).astype(int)

            # Count samples
            positive_count = y_binary.sum()
            negative_count = len(y_binary) - positive_count
            print(f"  Positive samples ({class_name}): {positive_count}")
            print(f"  Negative samples (others): {negative_count}")
            print(f"  Ratio: {positive_count/negative_count:.3f}")

            # Handle class imbalance with class_weight
            clf = RandomForestClassifier(
                **self.params,
                class_weight='balanced'  # Important for imbalanced data
            )
            clf.fit(X, y_binary)
            self.classifiers[class_name] = clf

        return self

    def predict_proba(self, X):
        """
        Get probability for each class
        """
        proba_matrix = np.zeros((X.shape[0], len(self.class_names)))

        for i, class_name in enumerate(self.class_names):
            # Get probability of belonging to this class
            proba = self.classifiers[class_name].predict_proba(X)[:, 1]
            proba_matrix[:, i] = proba

        # Normalize probabilities to sum to 1
        proba_matrix = proba_matrix / proba_matrix.sum(axis=1, keepdims=True)

        return proba_matrix

    def predict(self, X, method='probability'):
```

```
        """
        Predict class using voting or probability method

        Parameters:
        - method: 'probability' (highest proba) or 'voting' (majority vote)
        """
        if method == 'probability':
            # Method 1: Choose class with highest probability
            proba_matrix = self.predict_proba(X)
            predictions = np.argmax(proba_matrix, axis=1)
            return np.array([self.class_names[i] for i in predictions])

        elif method == 'voting':
            # Method 2: Binary decisions with majority vote
            votes = np.zeros((X.shape[0], len(self.class_names)))

            for i, class_name in enumerate(self.class_names):
                # Get binary prediction (0 or 1)
                pred_binary = self.classifiers[class_name].predict(X)
                votes[:, i] = pred_binary

            # Handle ties and no votes
            row_sums = votes.sum(axis=1)
            multiple_predictions = row_sums > 1
            no_predictions = row_sums == 0

            # For rows with multiple votes, use probability method
            if multiple_predictions.any():
                proba_matrix = self.predict_proba(X)
                for idx in np.where(multiple_predictions)[0]:
                    # Get classes that voted positive
                    voting_classes = np.where(votes[idx] == 1)[0]
                    # Choose highest probability among voting classes
                    best_class = voting_classes[np.argmax(proba_matrix[idx,␣
↪voting_classes])]
                    votes[idx] = 0
                    votes[idx, best_class] = 1

            # For rows with no votes, use highest probability
            if no_predictions.any():
                proba_matrix = self.predict_proba(X)
                for idx in np.where(no_predictions)[0]:
                    best_class = np.argmax(proba_matrix[idx])
                    votes[idx] = 0
                    votes[idx, best_class] = 1

            predictions = np.argmax(votes, axis=1)
```

```
            return np.array([self.class_names[i] for i in predictions])

    def predict_with_confidence(self, X):
        """
        Predict class with confidence score
        """
        proba_matrix = self.predict_proba(X)
        predictions = np.argmax(proba_matrix, axis=1)
        confidence = np.max(proba_matrix, axis=1)

        return {
            'predictions': np.array([self.class_names[i] for i in predictions]),
            'confidence': confidence,
            'probabilities': proba_matrix
        }

# Convert y_train and y_test to class names for binary classifiers
y_train_names = label_encoder.inverse_transform(y_train)
y_test_names = label_encoder.inverse_transform(y_test)

# Initialize and train binary Random Forest with voting
binary_rf = BinaryRandomForestWithVoting(
    n_estimators=100,
    max_depth=20,
    min_samples_split=10,
    min_samples_leaf=4,
    random_state=42
)

binary_rf.fit(X_train_scaled, y_train_names, class_names)

# Try both methods and pick the best
methods_to_try = ['probability', 'voting']
best_rf_binary = None
best_rf_binary_score = 0
best_method_name = ''

for method in methods_to_try:
    print(f"\n{'='*60}")
    print(f"Evaluating {method.upper()} method")
    print(f"{'='*60}")

    # Predict using current method
    rf_train_pred_names = binary_rf.predict(X_train_scaled, method=method)
    rf_test_pred_names = binary_rf.predict(X_test_scaled, method=method)

    # Convert back to encoded labels
```

```
    rf_train_pred = label_encoder.transform(rf_train_pred_names)
    rf_test_pred = label_encoder.transform(rf_test_pred_names)

    # Get probabilities
    rf_test_proba = binary_rf.predict_proba(X_test_scaled)

    # Calculate metrics
    rf_train_acc = accuracy_score(y_train_names, rf_train_pred_names)
    rf_test_acc = accuracy_score(y_test_names, rf_test_pred_names)
    rf_train_f1 = f1_score(y_train_names, rf_train_pred_names, average='macro',␣
 ↪labels=class_names)
    rf_test_f1 = f1_score(y_test_names, rf_test_pred_names, average='macro',␣
 ↪labels=class_names)

    print(f"\nPerformance with {method.upper()} method:")
    print(f"  Train Accuracy: {rf_train_acc:.4f}")
    print(f"  Test Accuracy:  {rf_test_acc:.4f}")
    print(f"  Overfitting Gap: {rf_train_acc - rf_test_acc:.4f}")
    print(f"  Train F1 (macro): {rf_train_f1:.4f}")
    print(f"  Test F1 (macro):  {rf_test_f1:.4f}")

    # Store best method
    if rf_test_f1 > best_rf_binary_score:
        best_rf_binary_score = rf_test_f1
        best_rf_binary = {
            'method': method,
            'train_pred': rf_train_pred,
            'test_pred': rf_test_pred,
            'train_proba': binary_rf.predict_proba(X_train_scaled),
            'test_proba': rf_test_proba,
            'train_acc': rf_train_acc,
            'test_acc': rf_test_acc,
            'train_f1': rf_train_f1,
            'test_f1': rf_test_f1
        }
        best_method_name = method

print(f"\n  Best binary RF method: {best_method_name.upper()}")
print(f"   Test F1 Score: {best_rf_binary['test_f1']:.4f}")

# Use the best method for final predictions
rf_train_pred = best_rf_binary['train_pred']
rf_test_pred = best_rf_binary['test_pred']
rf_train_proba = best_rf_binary['train_proba']
rf_test_proba = best_rf_binary['test_proba']
rf_train_acc = best_rf_binary['train_acc']
rf_test_acc = best_rf_binary['test_acc']
```

```
rf_train_f1 = best_rf_binary['train_f1']
rf_test_f1 = best_rf_binary['test_f1']

# LightGBM
print("\n" + "="*60)
print("LIGHTGBM - MULTICLASS")
print("="*60)

# LightGBM for multiclass
best_lgb = lgbm.LGBMClassifier(
    objective='multiclass',  # KEY: multiclass objective
    num_class=3,  # Number of classes
    learning_rate=0.09,
    max_depth=10,
    n_estimators=100,
    num_leaves=31,  # Key LightGBM parameter
    min_child_samples=20,
    subsample=0.8,
    colsample_bytree=0.8,
    reg_alpha=0.1,  # L1 regularization
    reg_lambda=0.1,  # L2 regularization
    random_state=42,
    n_jobs=-1
)

# Train with early stopping
best_lgb.fit(
    X_train_scaled, y_train,
    eval_set=[(X_test_scaled, y_test)],
    eval_metric='multi_logloss',
    callbacks=[lgbm.early_stopping(stopping_rounds=50, verbose=True)]
)

# Predictions
lgb_train_pred = best_lgb.predict(X_train_scaled)
lgb_test_pred = best_lgb.predict(X_test_scaled)
lgb_train_proba = best_lgb.predict_proba(X_train_scaled)
lgb_test_proba = best_lgb.predict_proba(X_test_scaled)

# Metrics for LGB
lgb_train_acc = accuracy_score(y_train, lgb_train_pred)
lgb_test_acc = accuracy_score(y_test, lgb_test_pred)
lgb_train_f1 = f1_score(y_train, lgb_train_pred, average='macro')
lgb_test_f1 = f1_score(y_test, lgb_test_pred, average='macro')

print(f"\nLightGBM Performance:")
print(f"  Train Accuracy: {lgb_train_acc:.4f}")
```

```
print(f"  Test Accuracy:  {lgb_test_acc:.4f}")
print(f"  Overfitting Gap: {lgb_train_acc - lgb_test_acc:.4f}")
print(f"  Train F1 (macro): {lgb_train_f1:.4f}")
print(f"  Test F1 (macro):  {lgb_test_f1:.4f}")
print(f"  Best iterations: {best_lgb.best_iteration_}")

# Comparison visualization

# Accuracy Comparison
fig, axes = plt.subplots(2, 3, figsize=(15, 10))

# Bar plot comparison
ax1 = axes[0, 0]
models = ['Random Forest', 'LightGBM']
train_acc = [rf_train_acc, lgb_train_acc]
test_acc = [rf_test_acc, lgb_test_acc]
x = np.arange(len(models))
width = 0.35
ax1.bar(x - width/2, train_acc, width, label='Train', color='skyblue')
ax1.bar(x + width/2, test_acc, width, label='Test', color='lightcoral')
ax1.set_ylabel('Accuracy')
ax1.set_title('Accuracy Comparison')
ax1.set_xticks(x)
ax1.set_xticklabels(models)
ax1.legend()
ax1.grid(True, alpha=0.3)

# F1 Score Comparison
ax2 = axes[0, 1]
train_f1 = [rf_train_f1, lgb_train_f1]
test_f1 = [rf_test_f1, lgb_test_f1]
ax2.bar(x - width/2, train_f1, width, label='Train', color='skyblue')
ax2.bar(x + width/2, test_f1, width, label='Test', color='lightcoral')
ax2.set_ylabel('F1 Score (macro)')
ax2.set_title('F1 Score Comparison')
ax2.set_xticks(x)
ax2.set_xticklabels(models)
ax2.legend()
ax2.grid(True, alpha=0.3)

# Overfitting Gap
ax3 = axes[0, 2]
overfit_gap_rf = rf_train_acc - rf_test_acc
overfit_gap_lgb = lgb_train_acc - lgb_test_acc
ax3.bar(models, [overfit_gap_rf, overfit_gap_lgb], color=['orange', 'green'])
ax3.set_ylabel('Overfitting Gap (Train - Test)')
ax3.set_title('Overfitting Comparison (smaller is better)')
```

```
ax3.axhline(y=0, color='black', linestyle='-', linewidth=0.5)
ax3.grid(True, alpha=0.3)

# Confusion Matrices - Random Forest
from sklearn.metrics import ConfusionMatrixDisplay
ConfusionMatrixDisplay.from_estimator(best_rf, X_test_scaled, y_test,
                                      display_labels=class_names,
                                      ax=axes[1, 0], cmap='Blues')
axes[1, 0].set_title('Random Forest - Confusion Matrix')

# Confusion Matrices - LightGBM
ConfusionMatrixDisplay.from_estimator(best_lgb, X_test_scaled, y_test,
                                      display_labels=class_names,
                                      ax=axes[1, 1], cmap='Greens')
axes[1, 1].set_title('LightGBM - Confusion Matrix')

# Per-class F1 Score comparison
ax6 = axes[1, 2]
rf_per_class_f1 = f1_score(y_test, rf_test_pred, average=None)
lgb_per_class_f1 = f1_score(y_test, lgb_test_pred, average=None)
x = np.arange(len(class_names))
ax6.bar(x - width/2, rf_per_class_f1, width, label='Random Forest',␣
 ↪color='skyblue')
ax6.bar(x + width/2, lgb_per_class_f1, width, label='LightGBM',␣
 ↪color='lightgreen')
ax6.set_ylabel('F1 Score')
ax6.set_title('Per-Class F1 Score Comparison')
ax6.set_xticks(x)
ax6.set_xticklabels(class_names, rotation=45)
ax6.legend()
ax6.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

# Metrics table
print("\n" + "="*80)
print("COMPARISON TABLE")
print("="*80)

comparison_data = {
    'Metric': ['Accuracy', 'F1 (macro)', 'Precision (macro)', 'Recall (macro)'],
    'RF Train': [
        accuracy_score(y_train, rf_train_pred),
        f1_score(y_train, rf_train_pred, average='macro'),
        precision_score(y_train, rf_train_pred, average='macro'),
        recall_score(y_train, rf_train_pred, average='macro')
```

```
    ],
    'RF Test': [
        accuracy_score(y_test, rf_test_pred),
        f1_score(y_test, rf_test_pred, average='macro'),
        precision_score(y_test, rf_test_pred, average='macro'),
        recall_score(y_test, rf_test_pred, average='macro')
    ],
    'LGB Train': [
        accuracy_score(y_train, lgb_train_pred),
        f1_score(y_train, lgb_train_pred, average='macro'),
        precision_score(y_train, lgb_train_pred, average='macro'),
        recall_score(y_train, lgb_train_pred, average='macro')
    ],
    'LGB Test': [
        accuracy_score(y_test, lgb_test_pred),
        f1_score(y_test, lgb_test_pred, average='macro'),
        precision_score(y_test, lgb_test_pred, average='macro'),
        recall_score(y_test, lgb_test_pred, average='macro')
    ]
}

comparison_df = pd.DataFrame(comparison_data)
print(comparison_df.to_string(index=False))

# Classification reports
print("\n" + "="*80)
print("RANDOM FOREST - CLASSIFICATION REPORT")
print("="*80)
print(classification_report(y_test, rf_test_pred, target_names=class_names))

print("\n" + "="*80)
print("LIGHTGBM - CLASSIFICATION REPORT")
print("="*80)
print(classification_report(y_test, lgb_test_pred, target_names=class_names))

# Feature importance comparison
fig, axes = plt.subplots(1, 2, figsize=(15, 8))

# Random Forest Feature Importance
rf_importance = best_rf.feature_importances_
rf_importance_df = pd.DataFrame({
    'feature': X.columns,
    'importance': rf_importance
}).sort_values('importance', ascending=True).tail(15)

axes[0].barh(rf_importance_df['feature'], rf_importance_df['importance'],␣
 ↪color='skyblue')
```

```
axes[0].set_xlabel('Importance')
axes[0].set_title('Random Forest - Top 15 Features')
axes[0].grid(True, alpha=0.3)

# LightGBM Feature Importance
lgb_importance = best_lgb.feature_importances_
lgb_importance_df = pd.DataFrame({
    'feature': X.columns,
    'importance': lgb_importance
}).sort_values('importance', ascending=True).tail(15)

axes[1].barh(lgb_importance_df['feature'], lgb_importance_df['importance'],␣
 ↪color='lightgreen')
axes[1].set_xlabel('Importance')
axes[1].set_title('LightGBM - Top 15 Features')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

# Final verdict
print("\n" + "="*80)
print("FINAL VERDICT")
print("="*80)

if lgb_test_f1 > rf_test_f1:
    print("LightGBM performs better on test data")
    print(f"   Improvement: {(lgb_test_f1 - rf_test_f1)*100:.2f}% higher F1␣
 ↪score")
elif rf_test_f1 > lgb_test_f1:
    print("Random Forest performs better on test data")
    print(f"{(rf_test_f1 - lgb_test_f1)*100:.2f}% higher F1 score")
else:
    print("Both models perform similarly")
```